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First Draft Final Report PowerPoint Slide

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1. Çayırova and Darıca District Shape file Matching Processing

- After I obtained all the necessary shape files from HDX [Humanitarian Data Exchange web sites \(humdata.org\)](https://humdata.org), I did the correlation (Matching Process) whether the shape files that I obtained from the two different sources are Similar or not,
- As a result, the HDX shape file has not matching exactly the Admin2 of Çayırova and Darıca. The updated date on the web posted by the HDX (OCHA) Information Management working group shows 22 March 2021 (it seems it is very recent and updated.
- But needs discussion to determine the correct Admin 2 shape file via Tele conference. Please see the next slide the two different sources separately and overlaying.

Çayırova and Darıca District



Overlay



WHO / MoH Source



HDX Source
Updated on 22 March 2021

Two Different Sources of Cayırova and Darıca administrative boundary shows slightly different shape file.

Presentation Outline

- Abstract
- Data Source Credibility issues,
- **Facilities**
 - 1. Population density methods
 - 2. Population Density classification
 - 3. Schools (n=134),
 - 4. Health Facilities (Clinics, Pharmacy, Health Center and Hospitals), (n=48)
 - 5. Industries(n=110),
 - 6. Buildings,(n=5770),
 - 7. Public Toilets(n=49),
 - 8. Shops(n=43),
 - 9. Supermarkets(n=29)
 - 10. Fuel Station,(n=26),
 - 11. Hairdresser(n=22)
 - 12. Car Part (n=8),
 - 12. Restaurants,(n=21),
 - 13. Car rental,(n=19),
 - 14. Food Court,(n=19),
 - 15. ATM,(n=18),
 - 16. Bakery,(n=16),
 - 17. Pharmacy,(n=15),
 - 18. Fast Food,(n=14),
 - 19. Pastry,(n=n=13),
 - 20. Café,(n=11),
 - 21. Bank,(n=10),
 - 22. Bucher,(n=10),
 - 23. Hotel,(n=26),
 - 24. Mall,(n=8),
 - 25. Store,(n=n=7),
 - 26. Dairy,(n=n=5),
 - 27. Post Office,(n=5),
 - 28. Kiosk,(n=4),
 - 29. Tailor,(n=4),
 - 30. Airport=2),
 - 31. Dormitory (n=7),
 - 32. Day care home (n=3)
 - 32. CORINE,

Presentation Outline (Con't...)

- Tea house (n=12)
- Town hall,(n=2),
- Stationery,(n=2),
- Place of Worship, (n=2),
- Market place, (n=2),
- Dry cleaning, (n=2),
- Chagrining Station, (n=2),
- Airports, (n=2),
- Veterinary,(n=1),
- Police Station, (n=1),
- Optician, (n=1),
- News agency, (n=1),
- Fire Station, (n=1),
- Farm, (n=1),
- Community Center, (n=1),

Summary

- Çayırova and Darıca District is located in the northern part of Turkey under the Kocaeli province. The two districts has a total of n=23 Mahalle localities. Of which n=9 of them are from Çayırova and n=14 from Darıca district. The two districts has a total area of 52,673 kilometer square. Of which Çayırova alone covers 28,982 square kilometer and Darıca covers 23,691. The total population from the two districts accounts n=355,069. Of which Çayırova alone contributes 140,273 and Darıca accounts 214,796. The age distribution from the two districts shows 15 (0-14) years accounts n=91,511 (26%), 15-64 n=245,248 (69%) and 65+ n=18,310 (5%). The gender distribution from the two districts shows n=214,796 (60%) are female and n=140,273 (40%) of them are Males. The two highly populated places are Cayirova at the Cayirova district and Danca place from Darıca districts. The high population density from the two districts shows at the Southern part of Cayirova and north-eastern part of Darıca. The average population density for the two districts cumulative shows 11.59 per square kilometer. Çayırova population density alone shows 10.75 per square kilometer and 13.06 per square kilometer at Darıca district. The two highly populated localities (Mahalle) are Ozgurluk from Cayirova, Kazimkarabekir and Zincilikuyu from Darıca and Kazimkarabekir and Zincilikuyu from Darıca district. A total of n=6613 facilities are being observed. Of which n=4684 (71%) from Darıca and n=1929 (29%) from Cayirova district. From the two districts, the major contribution of facilities shows buildings n=5770 followed by apartment n(n=265), Schools (n=134), Industry (n=110), Public Toilets (n=49), Health facilities (n=48) are the major observations.

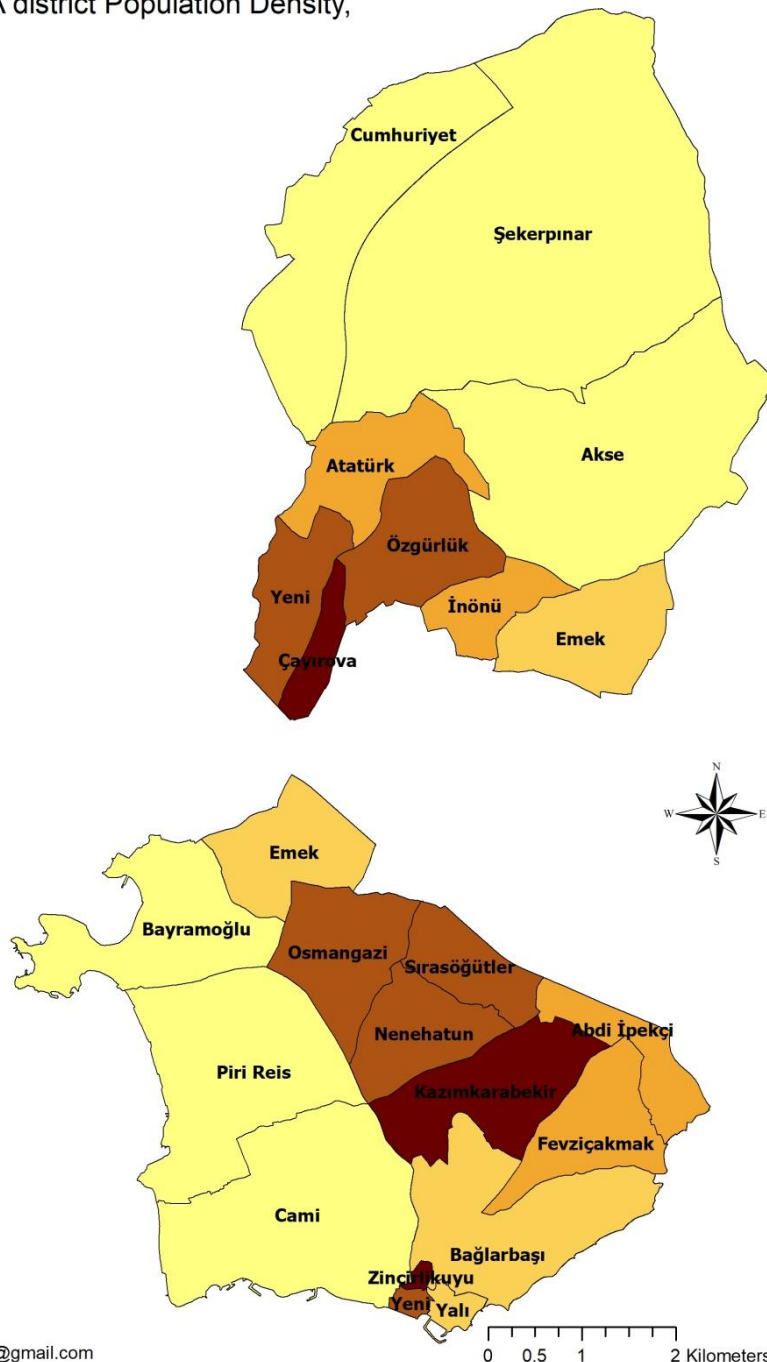
No	Indicator	Districts		Total
		Çayırova	Darıca	
1	Area Square Kilometer (Km2)	28,982 area sqkm2	23,691 aea sqkm2	52,673 area sqkm2
2	Total Population	140,273 (40%)	214,796 (60%)	355,069
3	Male	68,876 (49%)	108,791 (51%)	140,273 (40%)
4	Female	71,397 (51%)	106,055 (49%)	214,796 (60%)
5	<15 (0-14) Years	37,360 (27%)	54,151 (25%)	91,511 (26%)
6	15-64 years	97,036 (69%)	148,212 (69%)	245,248 (69%)
7	65+ years	5,877 (4%)	12,433 (6%)	18,310 (5%)
8	High Populated place	Zincirlikuyu	Özgürlük	Zincirlikuyu, Özgürlük
9	Population density	11.59/sqkm2	13.06/sqkm2	
10	High Population density place	Çayırova	Kazimkarabekir and Zincirlikuyu	Çayırova, Kazimkarabekir and Zincirlikuyu

No	Indicator	Districts		Total
		Çayirova	Darıca	
1	# of Hospital, n=4	1	3	4
2	# of Health Center, n=25	12	13	25
3	# of Pharmacy, n=15	3	12	15
5	# of Schools, n=134	54	80	134
6	# of Apartments, n=265	191	74	265
7	# of Airports, n=2	1	1	2
8	# of Buildings, n=5770	1669	4101	5770
9	# of Industrial places	102	35	137
10	# of Commercial Places	1	5	6
11	# of Hotels, n=26	6	20	26
12	# of Farms, n=6	0	6	6
13	# of Store(Warehouse),n=11	5	6	11
14	# of Garages	0	2	2
15	# of Green house,n=18	0	18	18
16	# of Dormitory	3	5	8
17	# of Gas Station (Fuel)	15	11	26
18	# of Public Toilets,n=49	1	48	49
19	# of Shops, n=43	0	43	43
20	# of Supermaret, n=29	3	26	29

No	Indicator	Districts		Total
		Çayırova	Darıca	
21	# of Restaurants, n=21	3	17	21
22	# of Car rental, n=19	9	10	19
23	# of Food Court, n=19	0	19	19
24	# of Bakery, n=16	3	13	16
25	# of Fast Food, n=14	3	11	14
26	# of Hairdresser, n=22	0	22	22
27	# of Shops(General), n=43	0	43	43
28	#of Pastry, n=13	1	12	13
29	# of Café, n=11	1	10	11
30	# of Bank, n=10	5	5	10
31	# of Buchery, n=10	0	10	10
32	# of Mall, n=8	0	8	8
33	# of Dairy, n=5	0	5	5
34	# of Post office, n=5	1	4	5
35	# of Kiosk, n=4	0	4	4
36	# Fire Station	0	1	1
37	# Market Place	1	1	2
38	# Community Center	0	1	1
39	# News agent	0	1	1
40	# of CORINE	40	22	62

Çayırova and DARICA district Population Density methods , Turkey, July, 2021

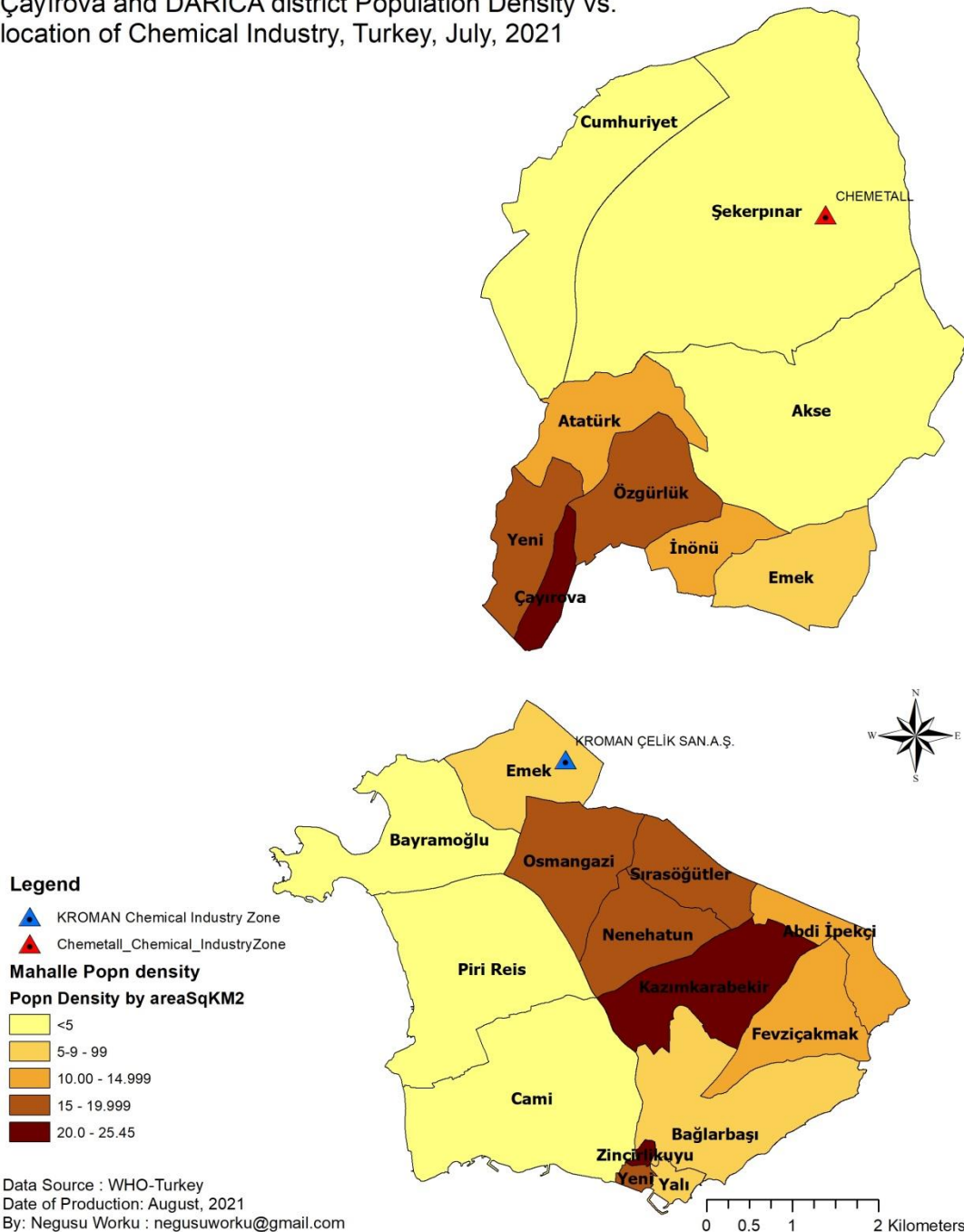
ilce_id	ilce_adi	mah_id	mah_adi	yuzolcum	tanitimkod	shape_Leng	shape_Area	Kadin_Nufu	Erkek_Nufu	Nufus_0_14	Nufus_65_u	Top_Nufus	areaMeter	AreaMeterK	AreaSqKM2
2059	Çayırova	50298	Emek	1733315	30	5742	1709084	5637	5845	3105	397	11482	1709084	1709	6.7
2060	Darica	50334	Nenehatun	1332147	19	5167	1332147	11954	12542	6941	1000	24496	1332147	1332	18.4
2060	Darica	50284	Sarımsaklılar	997727	1904	4606	997727	7526	7959	4247	698	15485	997727	998	15.5
2060	Darica	50332	Fevzi Paşakmak	1492710	17	6403	1492710	9656	9882	4592	1297	19538	1492710	1493	13.1
2060	Darica	50333	Kazım Karabekir	2030464	18	7576	2030464	20492	20985	10752	2148	41477	2030464	2030	20.4
2060	Darica	50338	Yeni	108641	23	1511	108641	935	900	385	210	1835	108641	109	16.8
2059	Çayırova	50317	Paşakerpemci	11361050	102	15160	11361050	1261	1403	642	150	2664	11361050	11361	0.2
2060	Darica	50337	Yalı	218485	22	3383	218485	564	590	200	119	1154	218485	218	5.3
2060	Darica	50336	Piri Reis	3706932	21	9742	3706932	4531	4783	2487	442	314	3706932	3707	0.1
2060	Darica	50329	Bayramoğlu	2479300	14	11620	2479300	3833	3941	1661	656	7774	2479300	2479	3.1
2060	Darica	50335	Osmangazi	1669592	20	5842	1669592	14614	15236	7813	1569	29850	1669592	1670	17.9
2060	Darica	50328	Bağlarbaşı	2862143	13	10476	2862143	13928	13736	6444	1836	27664	2862143	2862	9.7
2060	Darica	50267	Abdi Paşa	959084	1903	6168	959084	6365	6717	3132	851	13082	959084	959	13.6
2060	Darica	50331	Emek	1519829	16	5539	1519829	6270	6317	3067	834	12587	1519829	1520	8.3
2059	Çayırova	50296	Akse	6136074	28	12364	6136074	10488	10697	6123	625	21185	6136074	6136	3.5
2059	Çayırova	50297	Atatürk	1567662	29	7728	1567662	8975	9206	5203	684	18181	1567662	1568	11.6
2059	Çayırova	50290	Yeni	1221055	1901	5807	1221055	10017	10387	5105	1157	20404	1221055	1221	16.7
2060	Darica	50330	Cami	4253550	15	10826	4253550	4548	4465	2186	590	9013	4253550	4254	2.1
2059	Çayırova	50299	Çınarlı	847558	31	4647	847558	4698	4998	2616	399	9696	847558	848	11.4
2059	Çayırova	50300	Üzgeçirmen	1740754	32	6034	1740754	15788	16595	8975	1111	32383	1740754	1741	18.6
2059	Çayırova	50318	Cumhuriyet	3892255	103	11492	3892255	5742	5880	2578	677	11622	3892255	3892	3.0
2059	Çayırova	50273	Çayırova	506262	1902	3974	506262	6270	6386	3013	677	12656	506262	506	25.0
2060	Darica	50339	Zincirlikuyu	60145	24	1152	60145	789	738	244	183	1527	60145	60	25.5



Çayırova and Darıca District Population density findings

- * Low population density in the **northern, north-eastern** and **north-western** part of Çayırova District,
- * Low population density **half northern, half southern** and the **entire western part** of Darıca District,*High population density in the Southern part of Çayırova District,
- * High Population density in the north-eastern part of Darıca District,

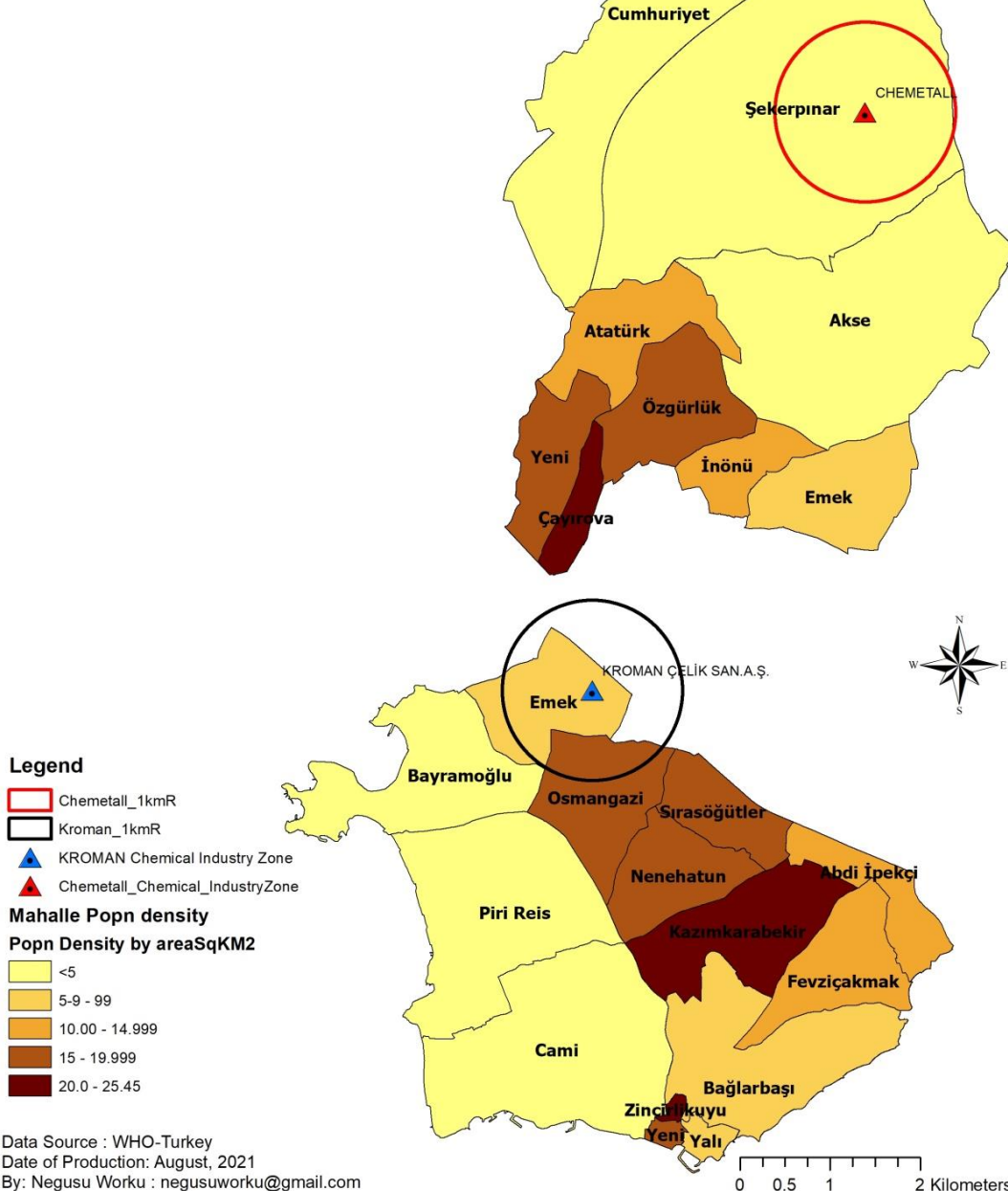
Çayırova and DARICA district Population Density vs. location of Chemical Industry, Turkey, July, 2021



1. Chemetall Chemical industry zone located in the north eastern part of Çayırova and

2.Kroman Chemical industry Zone located in the northern part of Darica.

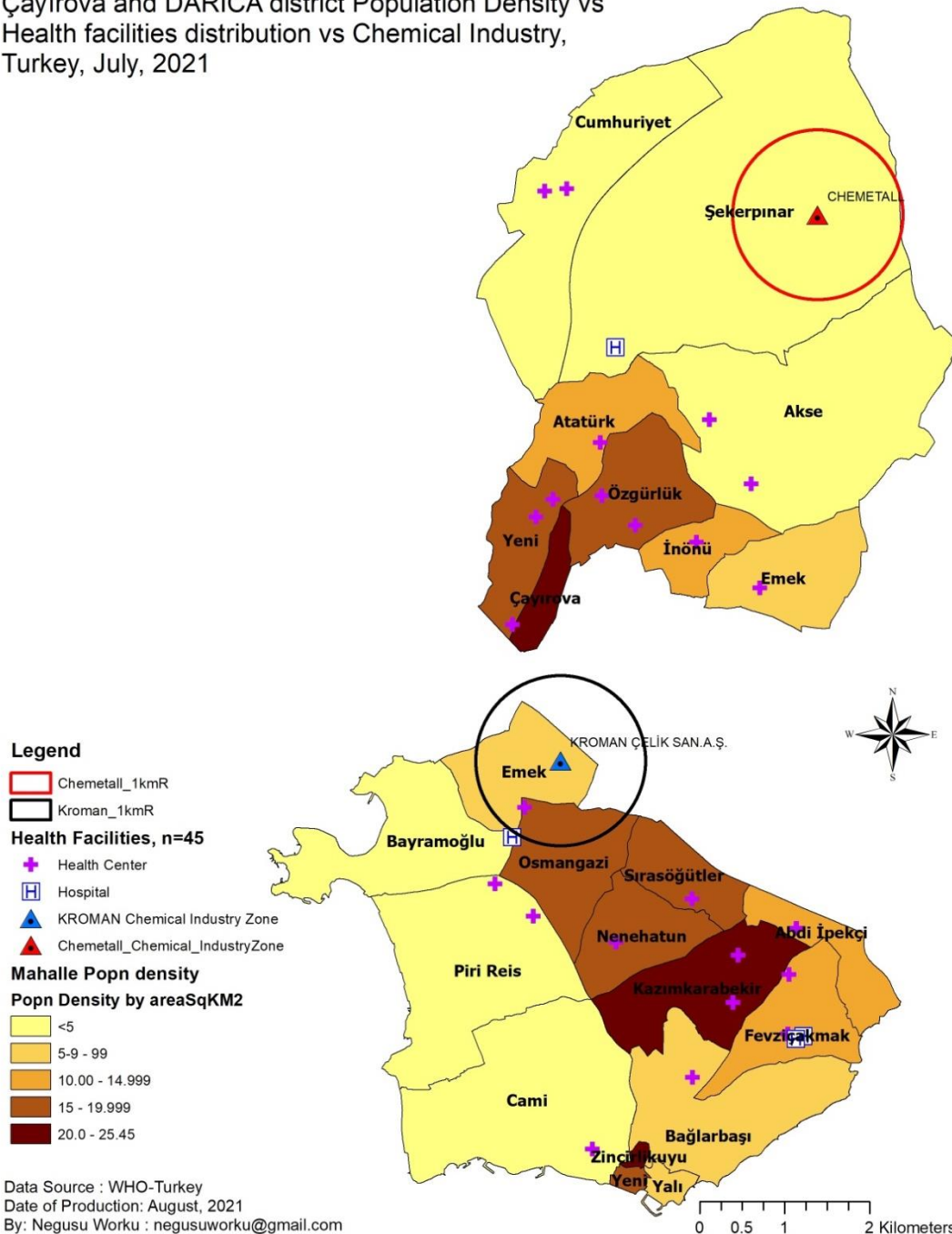
Çayırova and DARICA district Population Density vs the location of Chemical Industry vs. Kilometer radius, Turkey, July, 2021



Risk level within 1 km radius vs. population density

1. Çayırova within 1kmR population density is <5 person per square kilometer,

2. Darica district population density within 1 kmR shows from 5.9 – 19.99 Person per square kilometer,



Health Facilities (Health Center, Hospital and Pharmacy) shows n=44 Observations :

1. Health Center, n=25
2. Hospitals, n=4
3. Pharmacy, n=15

1. Çayırova district : health facilities availability 39% (n=1 hospital, n=12 Health Center and n=3 Hospitals),

2. Darica district : health facilities availability 61% (n=3 Hospitals, n=13 health centers, and n=11 Pharmacy),

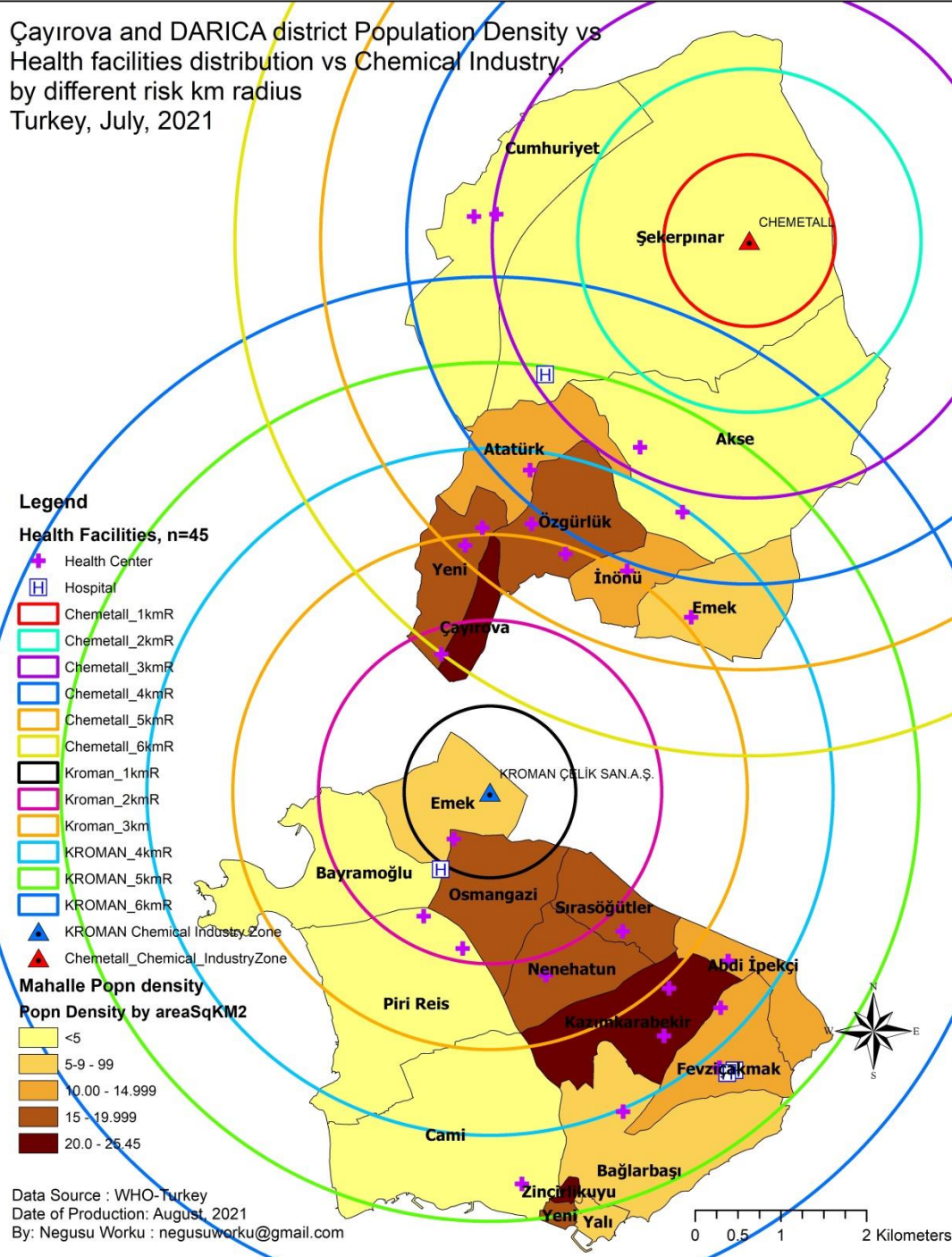
3. Çayırova district, within 1kmR no facilities located,

4. Darica, within 1 kmR three (3) health centers available,

5. Çayırova district , within 2kmR radius no Health facilities are available,

6. Darica, within 2kmR, n=10 health facilities are available, of which n=3 Health Centers, n=1 hospitals and n=6 Pharmacies are available.

Çayırova and DARICA district Population Density vs Health facilities distribution vs Chemical Industry, by different risk km radius Turkey, July, 2021



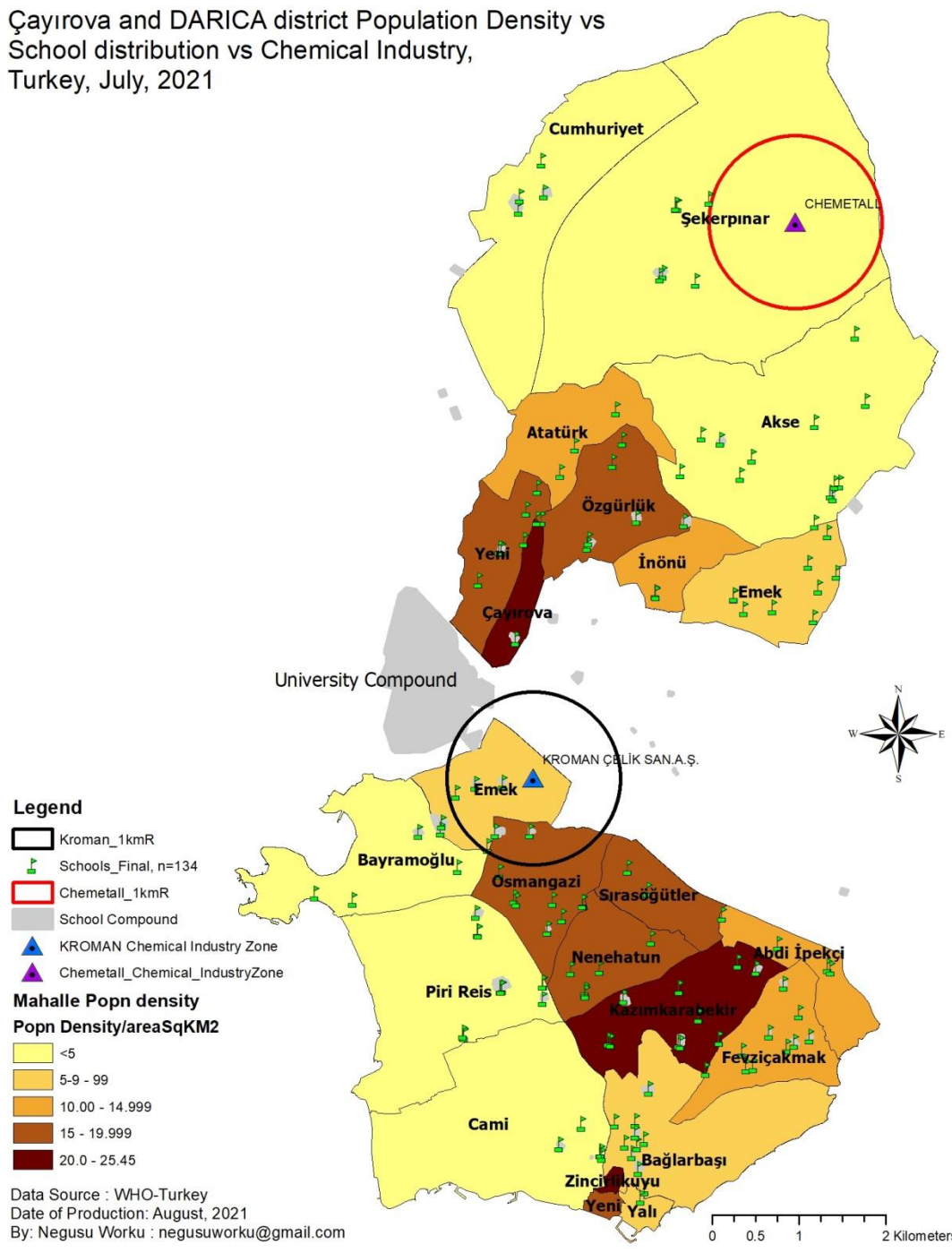
*From Kroman Chemical Industry **n=14** health facilities are available within 2kmR, of which **n=4 facilities** (1 health center and 3 pharmacies are available at CAYIROVA DISTRICT,

* From 3km radius from, Kroman Chemical industry, **n=19** Health facilities are observed, of which **n=6 of them were from Chemetall Chemical** industry (n=3 Pharmacy and n=3 Health Center,

**Cayırova and Darıca district Health Facilities risk level distribution from the chemical industry point radius, n=44
Health facilities, Turkey, August, 2021**

Disrict	Kilo Meter Radius	Health Facility Types					Risk level
		Pharmacy	Helth Center	Clinics	Hospital	List of HF's	
Darıca Kroman Industry Zone	1km radius (n=3)	2	1	0	0	..\HF_Final_kmR\Koroman_kmR1.xlsx	High
	2km radius (n=14)	9	4	0	1	..\HF_Final_kmR\Koroman_kmR2.xlsx	Medium
	3km radius (n=19)	10	8	0	1	..\HF_Final_kmR\Koroman_kmR3.xlsx	Low
	4km radius (n=30)	10	19	0	1	..\HF_Final_kmR\Koroman_kmR4.xlsx	
	5km radius (n=41)	14	23	0	4	..\HF_Final_kmR\Koroman_kmR5.xlsx	
	6km radius (n=41)	14	23	0	4	..\HF_Final_kmR\Koroman_kmR6.xlsx	Low
Total		14	26	1	4		
Disrict	Kilo Meter Radius	Health Facility Types					Risk level
		Pharmacy	Helth Center	Clinics	Hospital	List of HF's	
Cayırova Chemetall Industry Zon	1km radius (n=0)	0	0	0	0	No facility available within 1kmR	No Risk
	2km radius (n=0)	0	0	0	0	No facility available within 1kmR	No Risk
	3km radius (n=3)	0	2	0	1	..\HF_Final_kmR\Chemetall_kmR3.xlsx	Low
	4km radius (n=6)	0	5	0	1	..\HF_Final_kmR\Chemetall_kmR4.xlsx	Low
	5km radius (n=12)	0	11	0	1	..\HF_Final_kmR\Chemetall_kmR5.xlsx	
	6km radius (n=16)	4	11	0	1	..\HF_Final_kmR\Chemetall_kmR6.xlsx	
Total		2	11	0	1		

Çayırova and DARICA district Population Density vs
School distribution vs Chemical Industry,
Turkey, July, 2021



A total of n=134 Schools are available in the Cayirova and Darıca districts. Of which, n=37 Secondary Schools, n=33 Primary Schools,, n=2 middle schools n=28 Kindergarten and n=34 High Schools.

Moreover, four (n=4) Universities are located at the edge of the two districts boundary in the northern pick of Darıca district and the bottom edge of the Cayirova district within one (1kmR) kilometer radius.

Moreover, by different kilometer radius, from the Cayirova district, from the Point of Chemetall Chemical Industry, a total of n=9 Schools are located within 2km radius and from Darıca district, a total of n=30 Schools are at risk by 1 and 2kmR of which, n=3 kindergarten are among them.

School distribution by type by districts, n=134, Turkey, August, 2021

Districts	Kindergarten	Primary	Secondary	Middle	High	Total
Cayırova	11	13	17	0	13	54
Darİca	17	20	20	2	21	80
Total	38	33	37	2	34	134

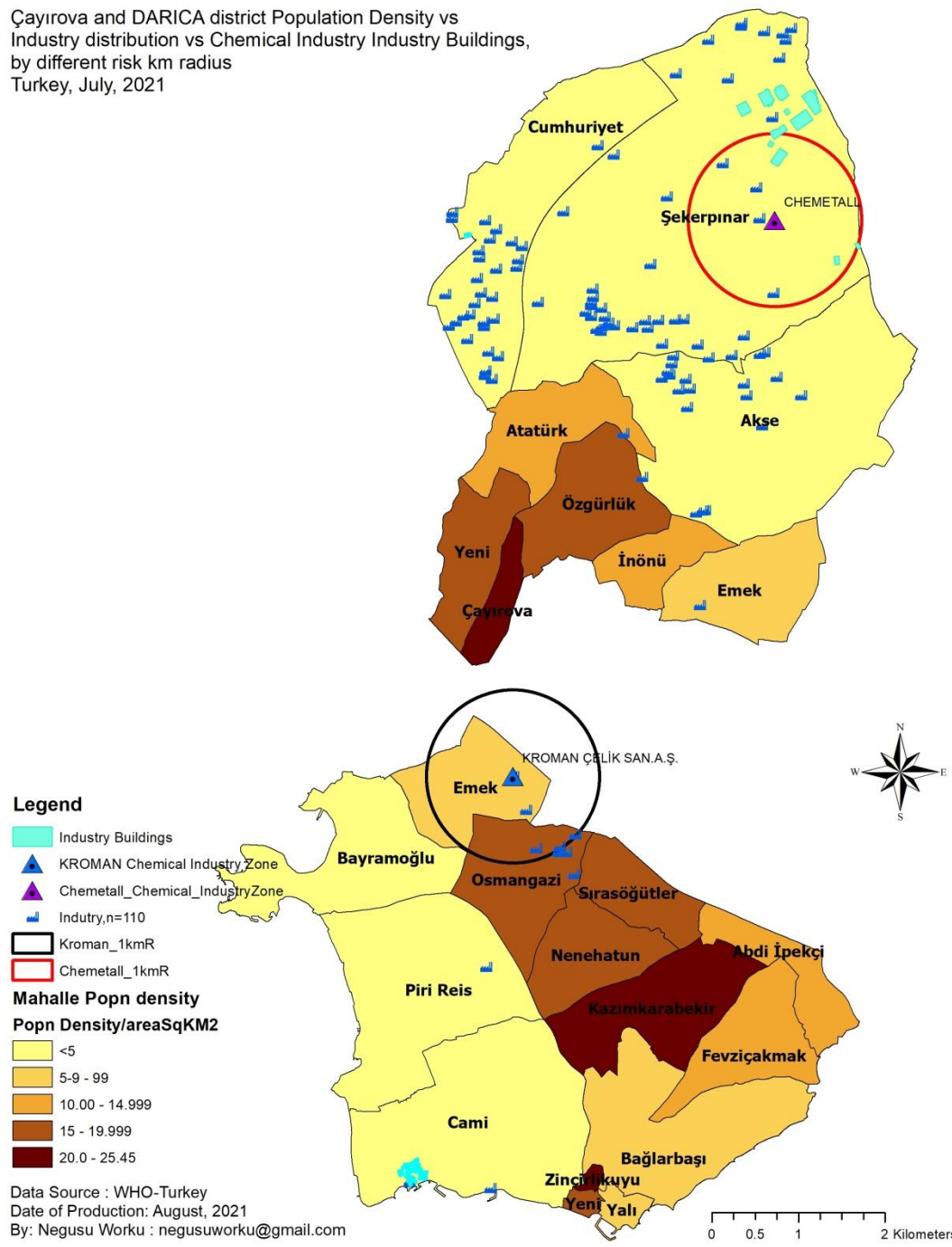
From the total n=134 School facilities, n=39 of them were found to be labeled as high risk category from the chemical industry zone. Of which n=9 of them are from Cayirova and n=30 of them are from Darİca. Risk level from 1km radius shows n=6 from Darica district including n=2 Primary, n=2 Secondary and n=2 high school. Those schools from Cayirova district located within 6km radius away from the Chemetall chemical zones, they became at risk from the other chemical site called Koroma chemical industry zone.

Cayırova Districts Summary of Schools distribution by risk level by different kilometer radius, n=54, Turkey, August, 2021

Schools	1kmR (n=0)	2kmR (n=9)	3kmR (n=17)	4kmR (n=34)	5kmR (n=50)	6kmR (n=54)	Total (n=134)	Risk Level	List of School Facilities by different KM Rdius
Kindergarten	0	3	4	8	11	11	37	High Risk	..\ListofSchoolsn134\CListofsk1kmR.xlsx
Primary School	0	1	1	6	11	13	32	Middle Risk	..\ListofSchoolsn134\CListofsk2kmR.xlsx
Secondary School	0	2	6	10	15	17	50	Low risk	..\ListofSchoolsn134\CListofsk3kmR.xlsx
Middle School	0	0	0	0	0	0	0	Low risk	..\ListofSchoolsn134\CListofsk4kmR.xlsx
High School	0	3	6	10	13	13	45	Low risk	..\ListofSchoolsn134\CListofsk5kmR.xlsx
Total	0	9	17	34	50	54	164		..\ListofSchoolsn134\CListofsk6kmR.xlsx

Darİca Districts Summary of Schools distribution by risk level by different kilometer radius, n=80, Turkey, August, 2021

Schools	1kmR (n=6)	2kmR (n=24)	3kmR (n=50)	4kmR (n=82)	5kmR (n=119)	6kmR (n=122)	Total (n=134)	Risk Level	List of School Facilities by different KM Rdius
Kindergarten	0	3	6	13	23	24	69	High risk	..\Asc134IIST\DListofsk1kmR.xlsx
Primary School	2	5	13	25	32	33	110	Middle risk	..\Asc134IIST\DListofsk2kmR.xlsx
Secondary School	2	9	14	24	33	33	115	Low risk	..\Asc134IIST\DListofsk3kmR.xlsx
Middle School	0	0	1	1	2	2	6	Low risk	..\Asc134IIST\DListofsk4kmR.xlsx
High School	2	7	15	19	9	30	82	Low risk	..\Asc134IIST\DListofsk5kmR.xlsx
Total	6	24	49	82	99	122	382		..\Asc134IIST\DListofsk6kmR2.xlsx



From the total $n=110$ industry sites, $n=98$ (90%) of them located in the Cayirova district and $n=11$ (10%) located in the Darıca district.

Moreover, 90% of the concentration of Industry factory concentrated in the Central and western part of Cayirova, where population density is <5 person per square kilometer.

In addition, Industrial buildings are also concentrated in the northern tip of Cayirova $n=13$ (35%) and in the Darıca district $n=24$ (70%) in the southern tip of Darıca district.

Those buildings located within 2kmR are from the Cayirova district along the Chemetall chemical industry Sites and the risk is very high as compare to those Industrial buildings located in the Darıca districts, where they are located within 6kmR.

Cayirova and Darİca Districts Summary of Industry Sites distribution by risk level by different kilometer radius, n=110, Turkey, August, 2021

District	Cayirova	Darİca	Total	Risk Level	Cayirova List	Darİca List
# of Industry Sites 1kmR	4	5	9	High	..\IndustryZoneAnalysis\Cayirova IndustrykmR Risk\CIndustry1kmR.xlsx	..\IndustryZoneAnalysis\Darİca IndustrykmR Risk\DIndustry1kmR.xlsx
# of Industry Sites 2kmR	24	8	32	High	..\IndustryZoneAnalysis\Cayirova IndustrykmR Risk\CIndustry2kmR.xlsx	..\IndustryZoneAnalysis\Darİca IndustrykmR Risk\DIndustry2kmR.xlsx
# of Industry Sites 3kmR	97	10	107	Medium	..\IndustryZoneAnalysis\Cayirova IndustrykmR Risk\CIndustry3kmR.xlsx	..\IndustryZoneAnalysis\Darİca IndustrykmR Risk\DIndustry3kmR.xlsx
# of Industry Sites 4kmR	97	14	111	Low	..\IndustryZoneAnalysis\Cayirova IndustrykmR Risk\CIndustry4kmR.xlsx	..\IndustryZoneAnalysis\Darİca IndustrykmR Risk\DIndustry4kmR.xlsx
# of Industry Sites 5kmR	98	28	126	Low	..\IndustryZoneAnalysis\Cayirova IndustrykmR Risk\CIndustry5kmRR.xlsx	..\IndustryZoneAnalysis\Darİca IndustrykmR Risk\DIndustry5kmR.xlsx
# of Industry Sites 6kmR	98	80	178	Low	..\IndustryZoneAnalysis\Cayirova IndustrykmR Risk\CIndustry6kmR.xlsx	..\IndustryZoneAnalysis\Darİca IndustrykmR Risk\DIndustry6kmR.xlsx

Çayırova and DARICA district Population Density vs
Hotel distribution vs by different risk km radius
(n=26 Hotels), Turkey, August, 2021

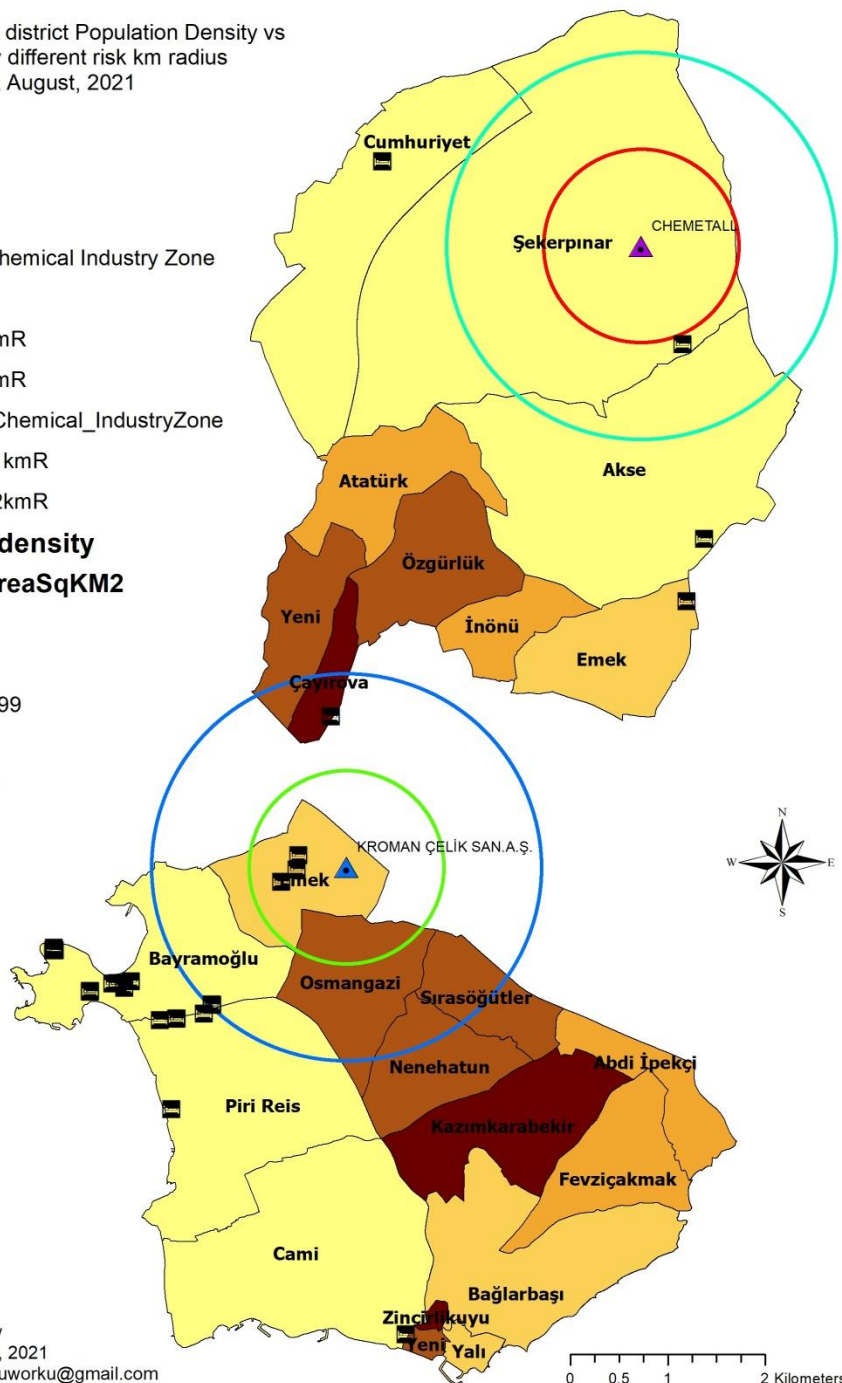
Legend

-  KROMAN Chemical Industry Zone
-  Hotel
-  Kroman_1kmR
-  Kroman_2kmR
-  Chemetall_Chemical_IndustryZone
-  Chemetall_1kmR
-  Chemetall_2kmR

Mahalle Popn density

Popn Density/areaSqKM2

-  <5
-  5-9 - 99
-  10.00 - 14.999
-  15 - 19.999
-  20.0 - 25.45



A total of n=26 Hotels are available
in the two districts,

Of which, n=20 (77%) from Darica
and

n=6 (23%) at Cayirova,

Within 1km radius from Kroman
chemical industry, only (n=3) Hotels
are available,

Within 1km radius from Chemetall
Chemical Industry, no hotels are
available,

Within 2km radius from Kroman
chemical industry, a total of n=5
Hotels are available of which 1 is
located at Cayirova district and 4 are
available at Darica,

Within 2km radius, from Chemetall
chemical industry, only n=1 Hotel is
available,

Cayirova and Darıca Districts Summary of Hotel Sites distribution by risk level by different kilometer radius, n=26, Turkey, August, 2021

KMRadius	District	Number of Hotel	Risk Level	List
1kmR	Cayirova	0	No Risk	..\Hotel_07082021\Choel1kmR.xlsx
2kmR	Cayirova	1	High Risk	..\Hotel_07082021\Choel2kmR.xlsx
3kmR	Cayirova	2	Medium risk	..\Hotel_07082021\Choel3kmRnew.xlsx
4kmR	Cayirova	4	Low risk	..\Hotel_07082021\Choel4kmR.xlsx
5kmR	Cayirova	4	Low risk	..\Hotel_07082021\Choel5kmR.xlsx
6kmR	Cayirova	5	Low risk	..\Hotel_07082021\Choel6kmR.xlsx
KMRadius	District	Number of Hotel	Risk Level	
1kmR	Darica	3	High Risk	..\Hotel_07082021\khoel1kmR.xlsx
2kmR	Darica	4	High Risk	..\Hotel_07082021\khoel2kmR.xlsx
3kmR	Darica	10	Medium risk	..\Hotel_07082021\khoel3kmR.xlsx
4kmR	Darica	17	Low risk	..\Hotel_07082021\khoel4kmR.xlsx
5kmR	Darica	21	Low risk	..\Hotel_07082021\khoel5kmR.xlsx
6kmR	Darica	33	Low risk	..\Hotel_07082021\khoel6kmR.xlsx

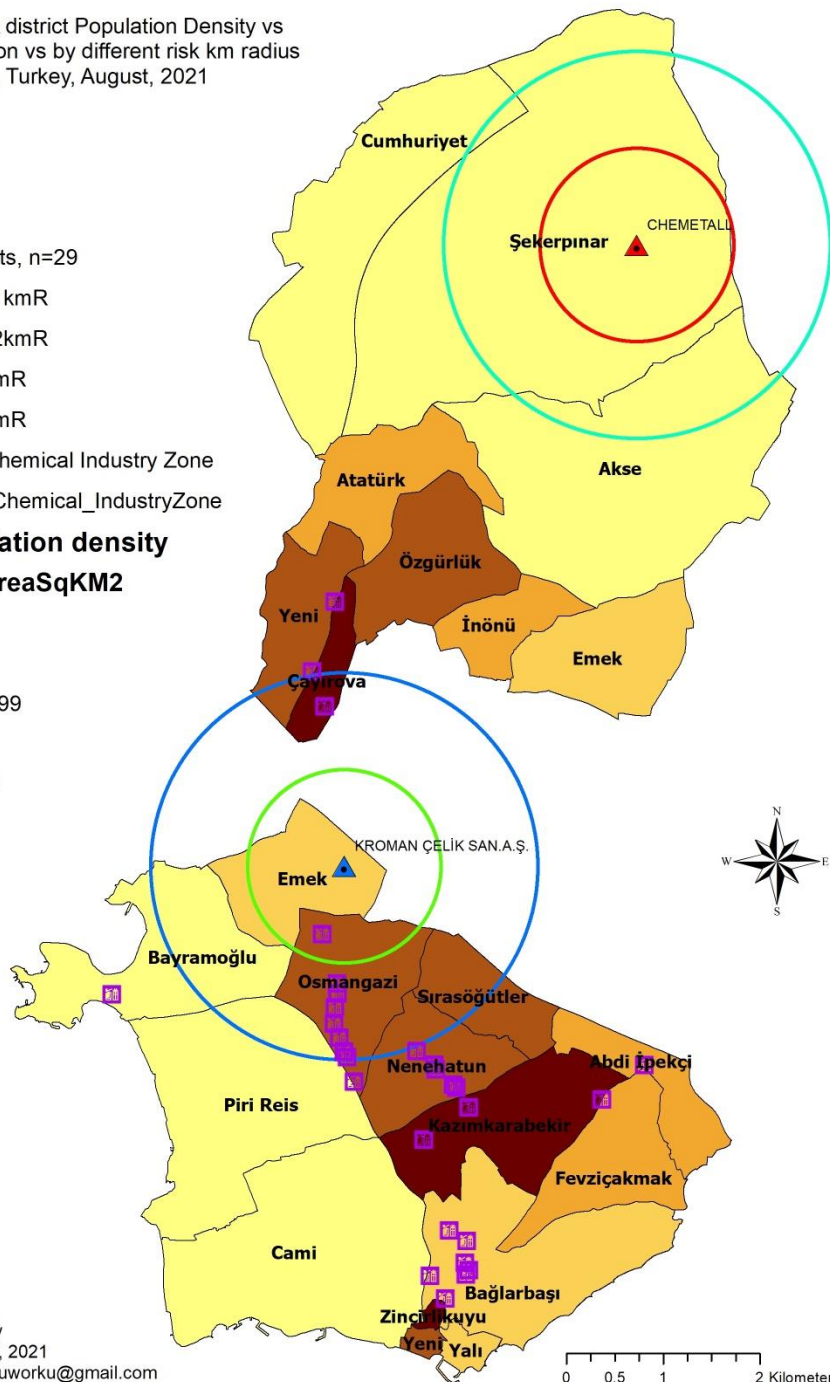
Legend

-  Supermarkets, n=29
-  Chemetall_1kmR
-  Chemetall_2kmR
-  Kroman_1kmR
-  Kroman_2kmR
-  KROMAN Chemical Industry Zone
-  Chemetall_Chemical_IndustryZone

Mahalle Population density

Popn Density/areaSqKM2

-  <5
-  5-9 - 99
-  10.00 - 14.999
-  15 - 19.999
-  20.0 - 25.45



A total of n=29 Supermarkets are available from the two districts. Of which n=3 (10%) from Cayirova and n=26 (90%) from Darica,

One (n=1) supermarkets available within 1kmR from the Kroman chemical industry, from Darica district,

From 1-4kmR from Cayirova district, 0 supermarkets available from the Chemetall Chemical industry,

Within 2kmR from Kroman Chemical Industry, a total of n=9 Supermarkets are available, of which n=1 from Cayirova districts,

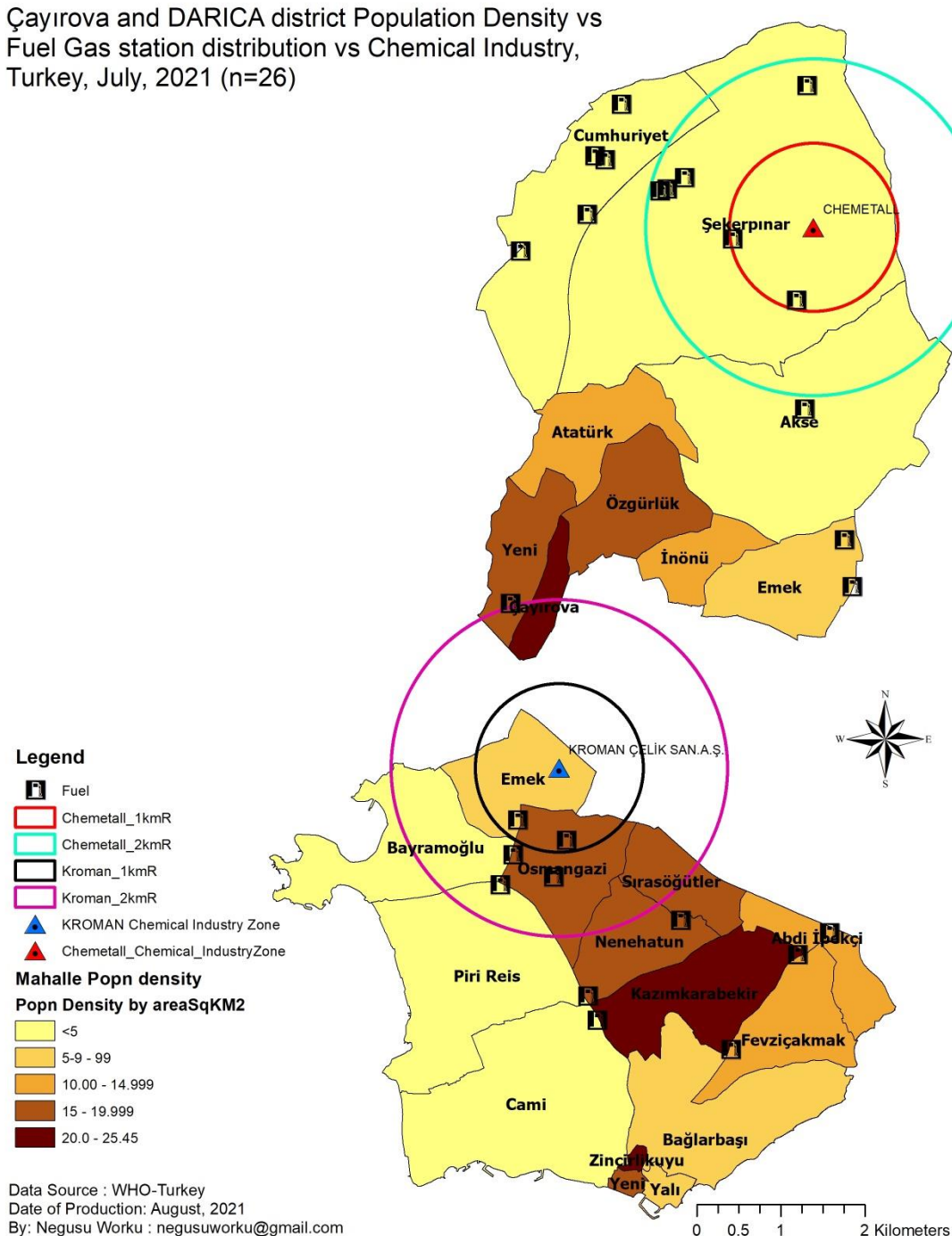
All supermarkets from Cayirova district, located 5-6 kmR from Chemetall industry zone, but these supermarkets are available within 2kmR from Kroman Chemical Industry from, Darica district,

Cayırova and Darıca Districts Summary of Supermarkets Sites distribution by risk level by different kilometer radius, n=29 Supermarkets, Turkey, August, 2021

KMR	Cayırova	Darıca	Total	Level of Risk	List of Supermarkets
1kmR	0	1	1	High	..\SuperM\DSUP1KMR.xlsx
2kmR	0	9	9	High	..\SuperM\DSUP2KMR.xlsx
3kmR	0	17	17	Medium	..\SuperM\DSUP3KMR.xlsx
4kmR	0	20	20	Low	..\SuperM\DSUP4KMR.xlsx
5kmR	1	29	30	Low	..\SuperM\DSUP5KMR.xlsx
6kmR	3	29	32	Low	..\SuperM\DSUP6KMR.xlsx

A total of n=10 Supermarkets are located in a high risk level between 1-2 km radius from the Kroman Chemical sites. No supermarkets are available in a high risk level at the Cayırova district. But those Supermarkets far from the Chemetall district in the southern part, they are at risk from the Kroman Chemical industry sites within 2-3 kilometer radius.

Çayırova and DARICA district Population Density vs
Fuel Gas station distribution vs Chemical Industry,
Turkey, July, 2021 (n=26)



Fuel Gas Station

A total of n=26 Fuel gas stations available,

15 at Cayirova,
11 Darica

A total of n=4 fuel gas stations available with 2kmR, 2 from Cayirova and n=2 from Darica,

Cayirova and Darİca Districts Summary of Fuel gas station Sites distribution by risk level by different kilometer radius, n=26, Turkey, August, 2021

KMRadius	District	Number of Hotel	Risk Level	List
1kmR	Cayirova	2	High Risk	
2kmR	Cayirova	6	High Risk	
3kmR	Cayirova	11	Medium risk	
4kmR	Cayirova	12	Low risk	
5kmR	Cayirova	13	Low risk	
6kmR	Cayirova	14	Low risk	
KMRadius	District	Number of Hotel	Risk Level	
1kmR	Darİca	2	High Risk	E:\FuelGas070821\FuelKromanKMR\K1kmRFuel.xlsx
2kmR	Darİca	5	High Risk	E:\FuelGas070821\FuelKromanKMR\K2kmRFuel.xlsx
3kmR	Darİca	8	Medium risk	E:\FuelGas070821\FuelKromanKMR\K3kmRFuel.xlsx
4kmR	Darİca	9	Low risk	E:\FuelGas070821\FuelKromanKMR\K4kmRFuel.xlsx
5kmR	Darİca	12	Low risk	E:\FuelGas070821\FuelKromanKMR\K5kmRFuel.xlsx
6kmR	Darİca	15	Low risk	E:\FuelGas070821\FuelKromanKMR\K6kmRFuel.xlsx

A total of n=26 Fuel gas stations are available in the two districts. Of Which n=11 (42%) are from Darİca and n=15 (58%) from Cayirova districts. Total n=4 Fuel gas stations are located within 1 km radius two from Cayirova and two from Darİca districts.

Çayırova and DARICA district Population Density vs Food, Bakery, Pastry... vs Chemical Industry, Turkey, July, 2021 (n=135)

Legend

Food, Cafe, Pastry....

- ◆ Bakery(n=16)
- Butcher (n=10)
- ☐ Cafe (n=11)
- ▼ Dairy (n=5)
- ◆ Fast_Food (n=14)
- ⊙ Food_Court (19)
- ☐ Hotel (n=26)
- ☐ Pastry (n=13)
- ☐ Restaurant (n=21)

Chemetall_1kmR

Chemetall_2kmR

Kroman_1kmR

Kroman_2kmR

KROMAN Chemical Industry Zone

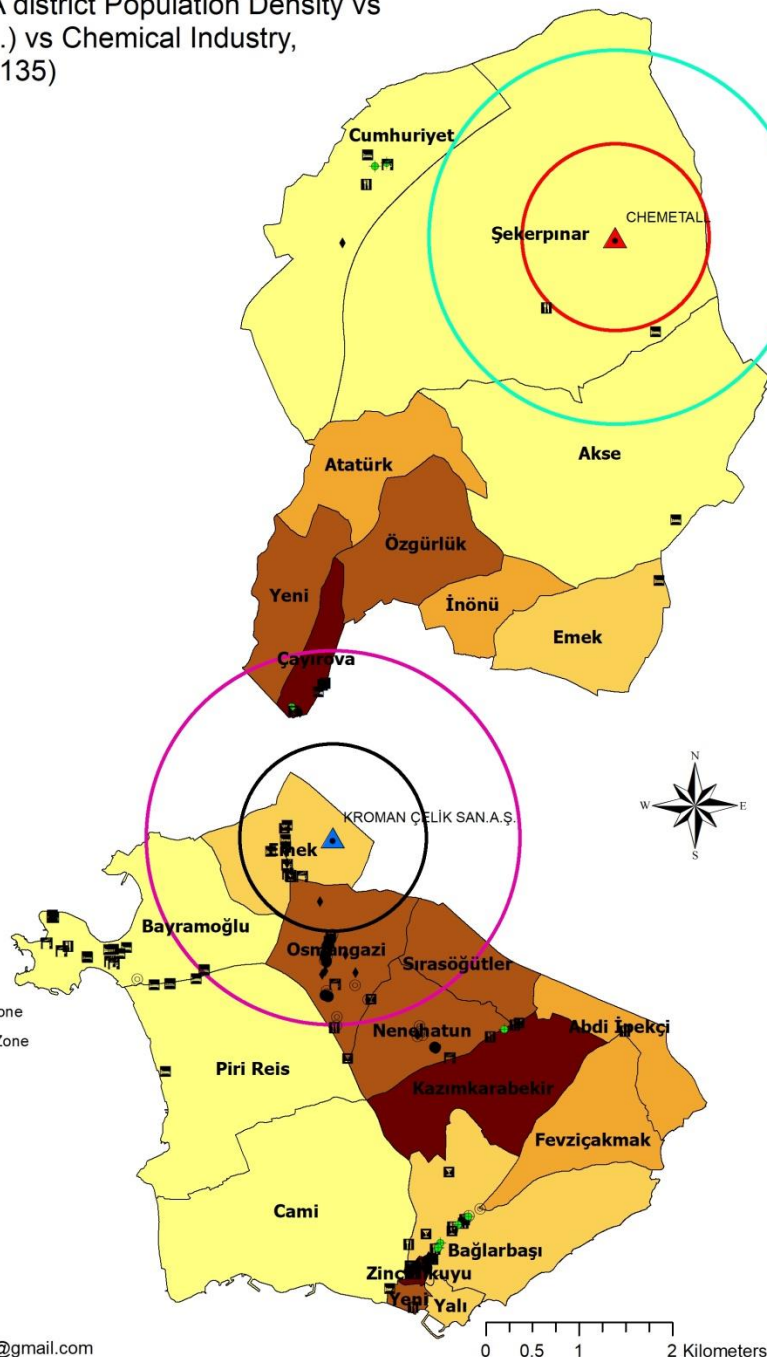
Chemetall_Chemical_IndustryZone

Mahalle Popn density

Popn Density by areaSqKM2

- <5
- 5-9 - 99
- 10.00 - 14.999
- 15 - 19.999
- 20.0 - 25.45

Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com



A total of n=135 facilities (including: hotel, restaurant, Pastry, Food court, Bakery, Fast food, Café, Dairy and Butcher),

n=18 (13%) facilities available at Cayirova, n=117 (87%) at Darica,

Within 1 kmR from Chemetall chemical industry from Cayirova district, no facility is available,

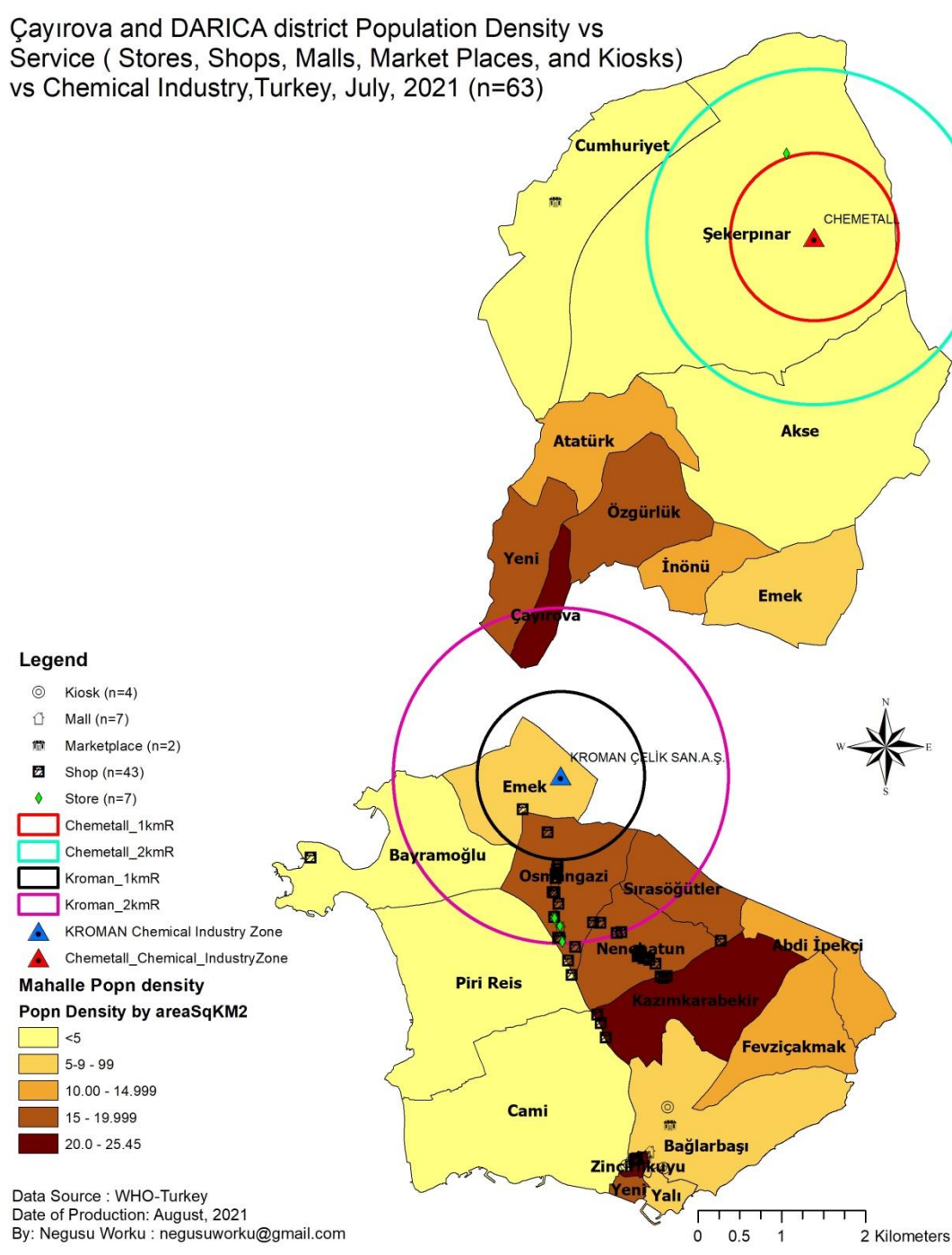
Within 1 kmR from Darica, Kroman Chemical industry, a total of n=14 facilities are at risk. Of which n=4 Bakery, n=4 Café, n=3 Hotel, n=2 Pastry and n=1 restaurant,

Within 2kmR from Darica, Kroman Chemical Industry Zone, a total of n=50 facilities are available of which n=7 of them are from Cayirova district, Chemetall industry,

Cayırova and Darıca Districts Summary of Hotel, Restaurant, Food Court, Bakery, Fast Food, Pastry, Butchery, Café and Dairy Sites distribution by risk level by different kilometer radius, n=135, Turkey, August, 2021

KMRadius	District	Number of Hotel	Risk Level	List of Facilities
1kmR	Cayırova	0	No Risk	
2kmR	Cayırova	2	High Risk	E:\n135\n135ca\CFood2kmRNew.xlsx
3kmR	Cayırova	9	Medium risk	
4kmR	Cayırova	10	Low risk	
5kmR	Cayırova	10	Low risk	
6kmR	Cayırova	14	Low risk	
>6kmR	Cayırova	17		
KMRadius	District	Number of Hotel	Risk Level	
1kmR	Darıca	14	High Risk	E:\n135\n135ca\KFood1kmR.xlsx
2kmR	Darıca	48	High Risk	E:\n135\n135ca\KFood2kmR.xlsx
3kmR	Darıca	65	Medium risk	E:\n135\n135ca\kFood3kmR.xlsx
4kmR	Darıca	81	Low risk	E:\n135\n135ca\kFood4kmR.xlsx
5kmR	Darıca	123	Low risk	E:\n135\n135ca\kFood5kmR.xlsx
6kmR	Darıca	126	Low risk	E:\n135\n135ca\kFood6kmR.xlsx

Çayırova and DARICA district Population Density vs Service (Stores, Shops, Malls, Market Places, and Kiosks) vs Chemical Industry,Turkey, July, 2021 (n=63)



A total of 63 Similar facilities are available.

Of which

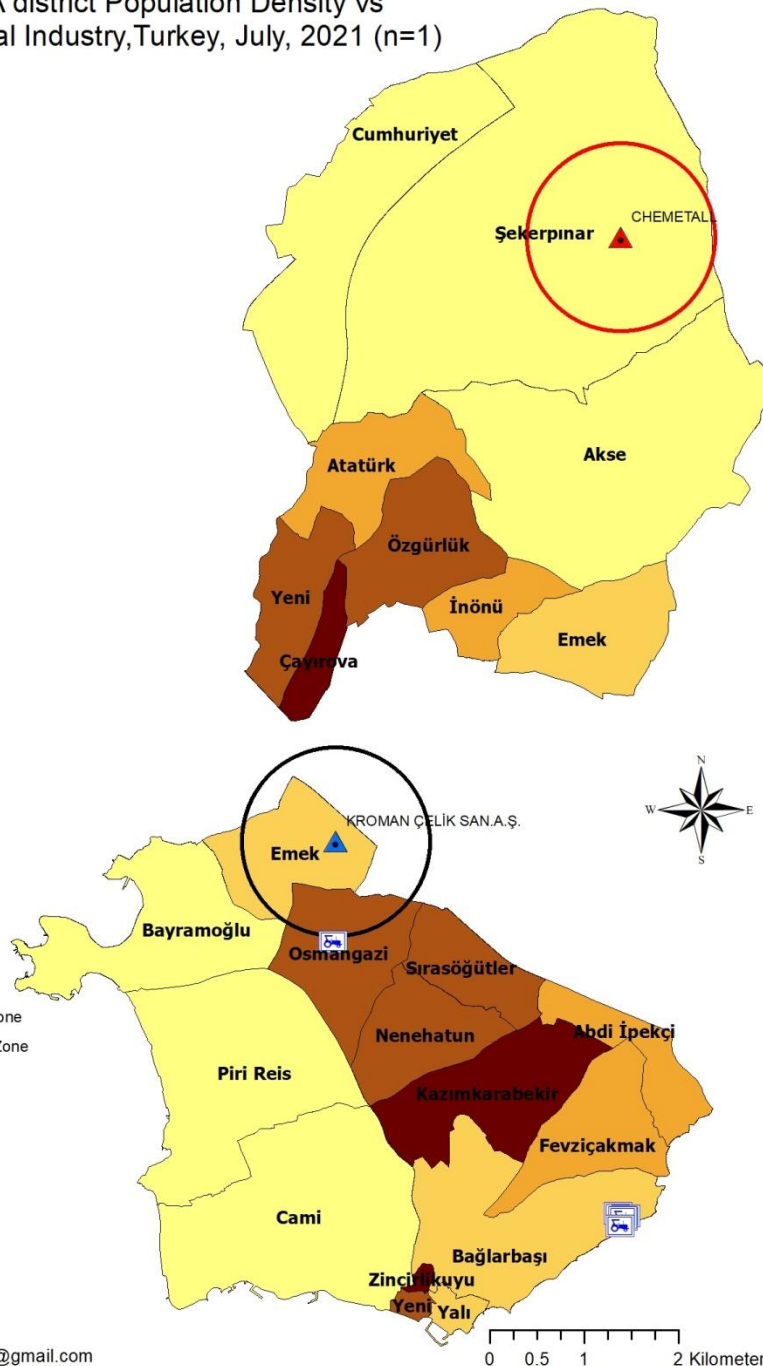
n=4 (6%) **kiosk**,
n=7 (11%) **Mall**,
n=43 (68%) **Shops**,
n=7 (7%) **Stores**

n=2 (3%) of them are from Cayirova,
n=61 (97%) from Darica,

n=2 Shops from Kroman chemical industry, from Darica district are located within 1kmR and

n=1 store located within 1kmR from the Chemetall Chemical industry from Cayirova district

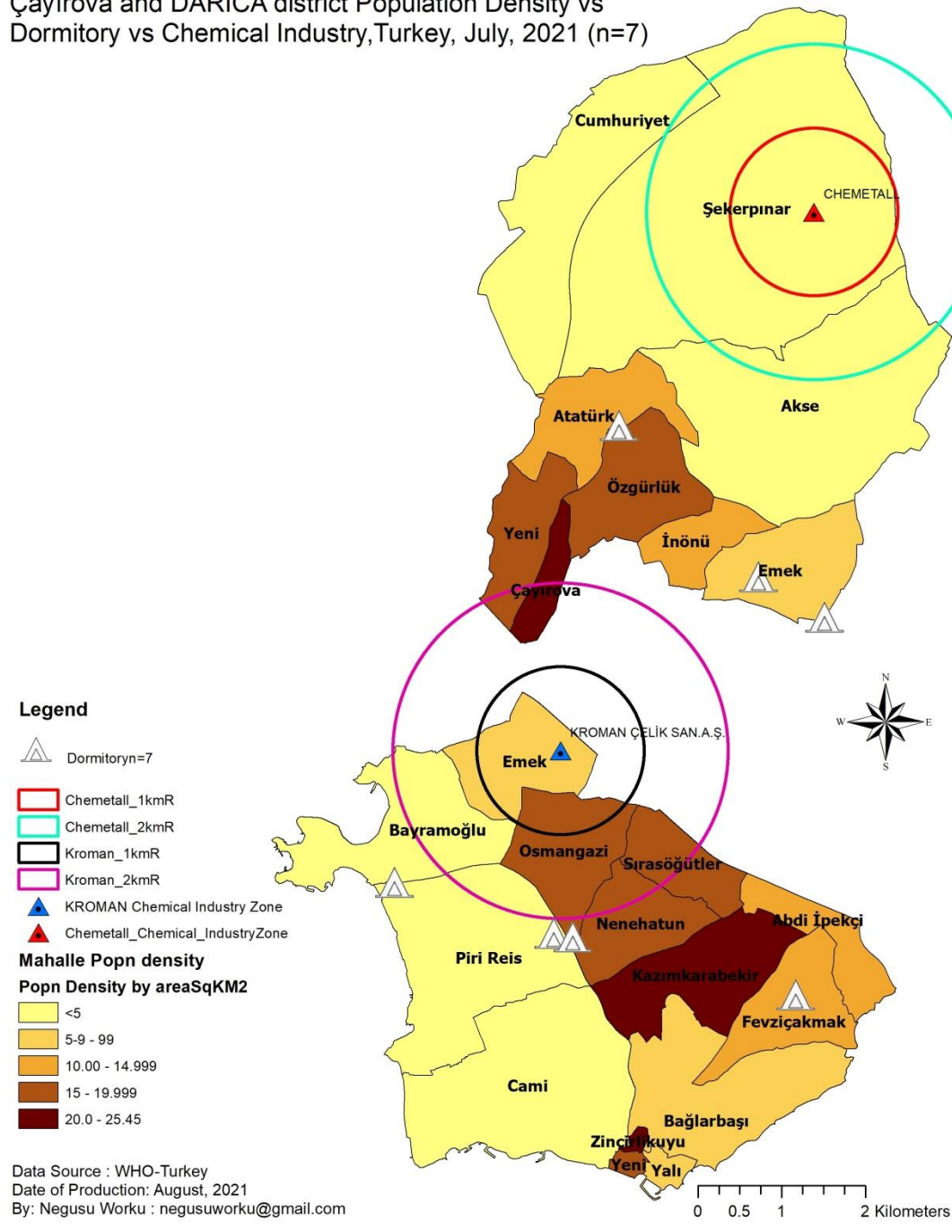
Çayırova and DARICA district Population Density vs
Farm area vs Chemical Industry,Turkey, July, 2021 (n=1)



n=6 Farm areas available in Darica district, of which only n=1 farm area available within 1kmR from the Kroman chemical industry,

n=5 Farm area are located within 5kmR from the Kroman Chemical industry, from Darica district,

Çayırova and DARICA district Population Density vs
Dormitory vs Chemical Industry,Turkey, July, 2021 (n=7)



Dormitory n=7 observations are located 2km radius from Kroman Chemical Industry and population density is high at Daricia District.

No Dormitory near to Chemetall Chemical Industry

Çayırova and DARICA district Population Density vs
Services vs Chemical Industry,Turkey, July, 2021 (n=115)

Legend

-  Airport
-  KROMAN Chemical Industry Zone
-  Chemetall_Chemical_IndustryZone
-  Tea House=12
-  Townhall=2
-  Stationery=2
-  Dry_Cleaning, n=2
-  Charging_Stationn=2
-  Veterinary, n=1
-  Police_Station=1
-  Optician=1
-  News_Agent=1
-  Fire Station=1
-  Community Center=1
-  Post_Office=5
-  Public_Toilet=49
-  Day Care Home=3
-  Bank=10
-  ATM_Machine=18

Chemetall_1kmR

Chemetall_2kmR

Kroman_1kmR

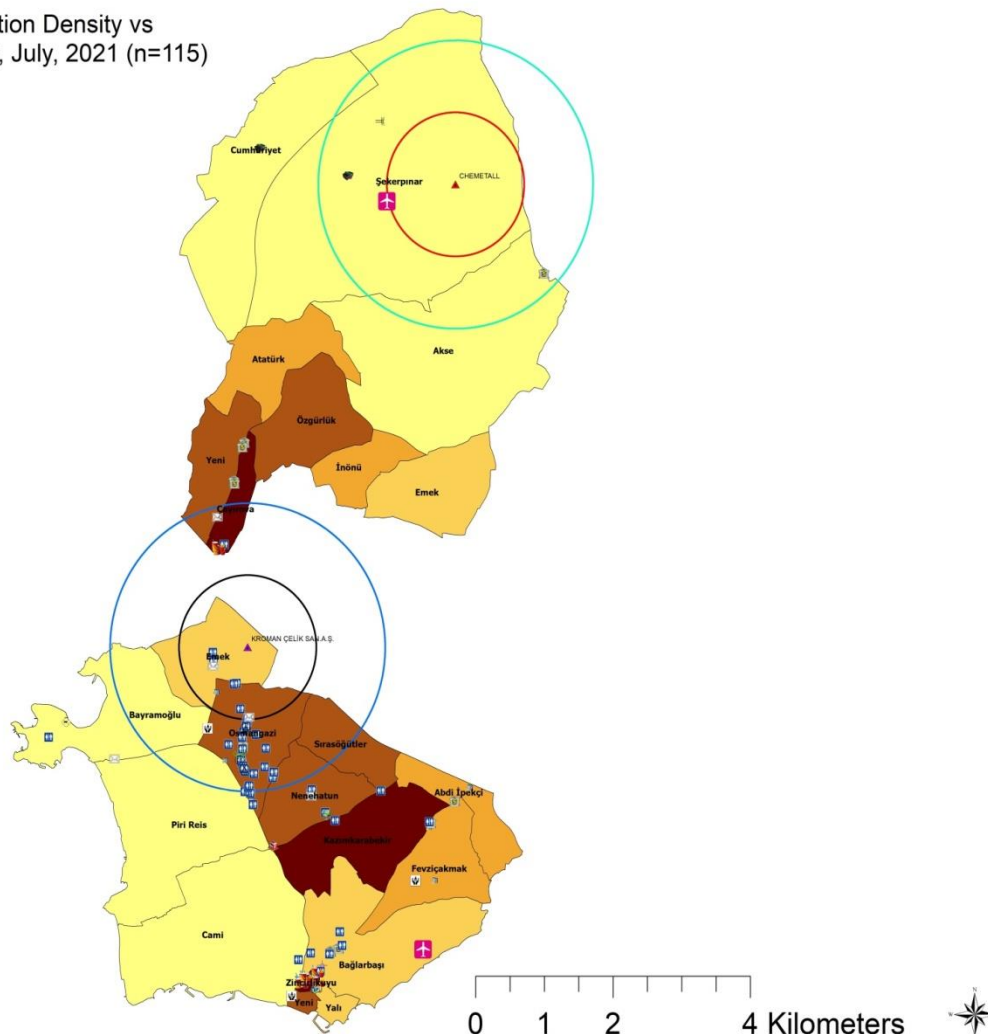
Kroman_2kmR

Mahalle Popn density

Popn Density by areaSqKM2

-  <5
-  5-9 - 99
-  10.00 - 14.999
-  15 - 19.999
-  20.0 - 25.45

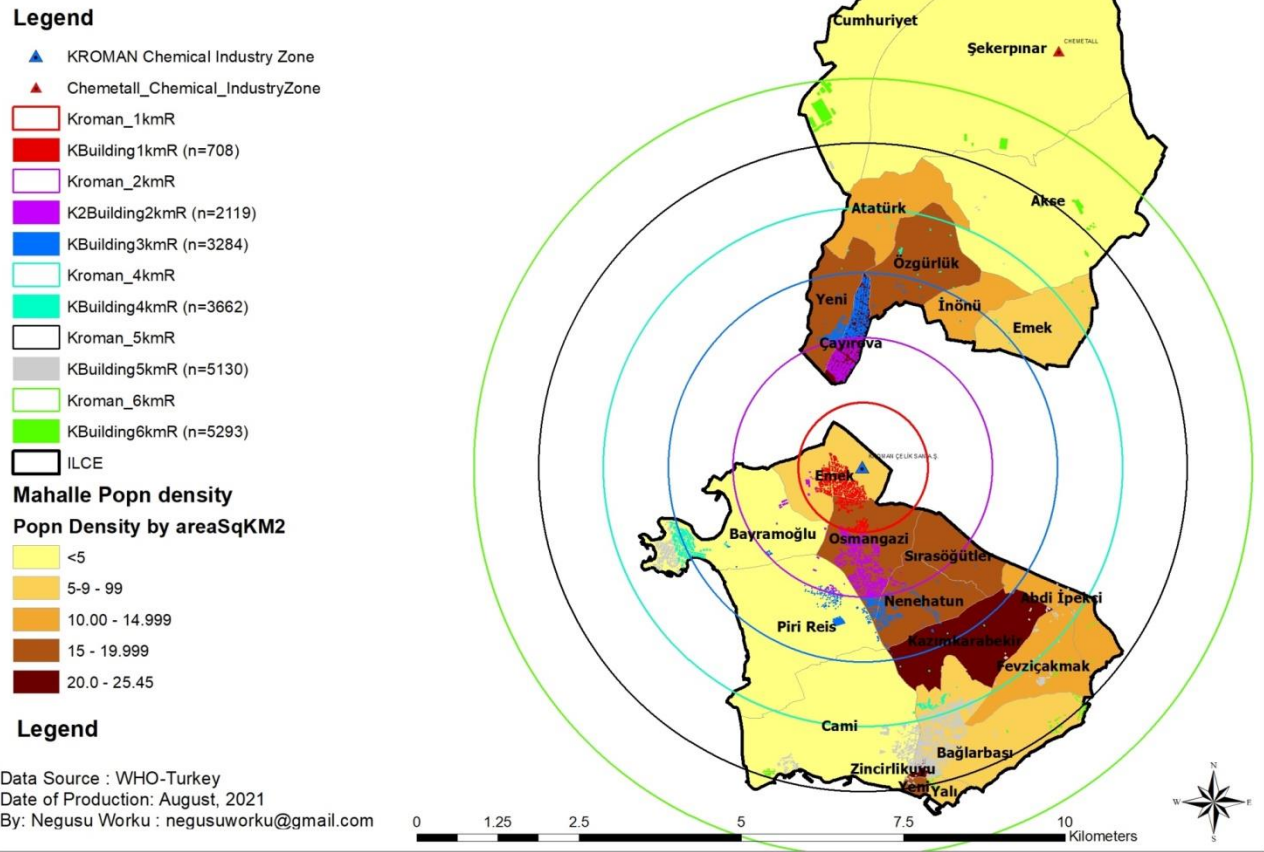
Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com



From n=115 Services
n=14(12%) are from
Cayırova, n=99(88%)
from Darica,

n=10 facilities are
available within 1kmR
including n=6 Public
Toilet, n=2 post offices
and n=2 ATM
machines,

Çayırova and DARICA district Population Density vs.
Buildings risk from the Kroman Chemical Industry, n=4101
Buildings, Turkey, August, 2021



A total of n=5770 buildings are available in both districts. Of which n=4101 buildings are available. Of Which n=4101 buildings are available. Of which n=708 of them were found to be within 1kmR. The majority of the services are high risk.

The majority of n=649 are unclassified buildings,
n=52 houses
n=4 Schools
n=1 Hotel,
n=1 Commercial buildings,

Cayırova and Darıca Districts Distribution of buildings by risk level by different kilometer radius, n=5770, Turkey, August, 2021

KMRadius	District	Number of Buildings	Risk Level	List of Facilities
1kmR	Cayırova	205	High Risk	E:\Mega_Report\Buildings_5770_DBF\Building1kmR.xlsx
2kmR	Cayırova	343	Medium risk	E:\Mega_Report\Buildings_5770_DBF\Building2kmR.xlsx
3kmR	Cayırova	404	Low risk	E:\Mega_Report\Buildings_5770_DBF\Building3kmR.xlsx
4kmR	Cayırova	448	Low risk	E:\Mega_Report\Buildings_5770_DBF\Building4kmR.xlsx
5kmR	Cayırova	541	Low risk	E:\Mega_Report\Buildings_5770_DBF\Building5kmR.xlsx
6kmR	Cayırova	993	Low risk	E:\Mega_Report\Buildings_5770_DBF\Building6kmR.xlsx
>6kmR	Cayırova	1669	Low risk	E:\Mega_Report\Buildings_5770_DBF\Building6p5kmR.xlsx
KMRadius	District	Number of Buildings	Risk Level	
1kmR	Darıca	708	High Risk	E:\Mega_Report\Buildings_5770_DBF\KBuilding1kmR.xlsx
2kmR	Darıca	2119	Medium risk	E:\Mega_Report\Buildings_5770_DBF\K2Building2kmR.xlsx
3kmR	Darıca	3284	Low risk	E:\Mega_Report\Buildings_5770_DBF\KBuilding3kmR.xlsx
4kmR	Darıca	3662	Low risk	E:\Mega_Report\Buildings_5770_DBF\kBuildings4kmR.xlsx
5kmR	Darıca	5130	Low risk	E:\Mega_Report\Buildings_5770_DBF\KBuilding5kmR.xlsx
6kmR	Darıca	5293	Low risk	E:\Mega_Report\Buildings_5770_DBF\KBuilding6kmR.xlsx

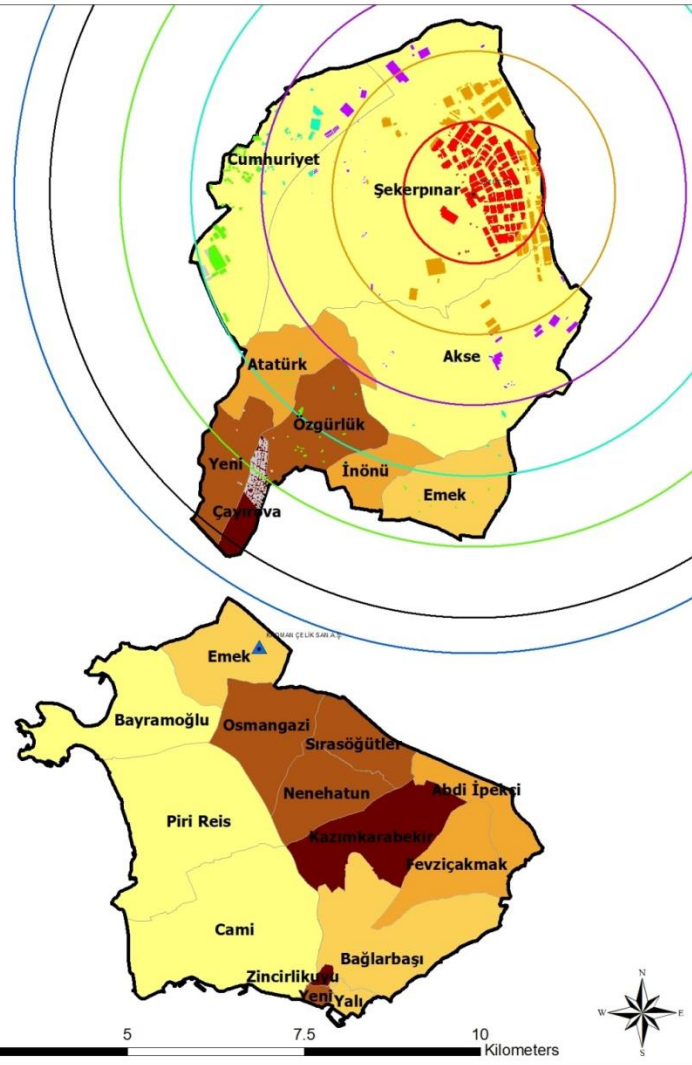
Çayırova and DARICA district Population Density vs. Buildings risk from the Chemetall Industry, n=1669 Buildings,Turkey, August, 2021

Legend

- ▲ KROMAN Chemical Industry Zone
- ▲ Chemetall_Chemical_IndustryZone
- Chemetall_1kmR
- CBuilding1kmR (n=205)
- Chemetall_2kmR
- CBuilding2kmR (n=343)
- Chemetall_3kmR
- CBuilding3kmR (n=404)
- Chemetall_4kmR
- CBuilding4kmR (n=448)
- Chemetall_5kmR
- CBuilding5kmR (n=541)
- Chemetall_6kmR
- CBuilding6kmR (n=993)
- Chemetall_6.5KMR
- ILCE

- Mahalle Popn density**
- Popn Density by areaSqKM2**
- <5
 - 5-9 - 99
 - 10.00 - 14.999
 - 15 - 19.999
 - 20.0 - 25.45

Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com



A total of n=5770 buildings are available in both districts. Of which n=1669 buildings are available in the Cayirova district.

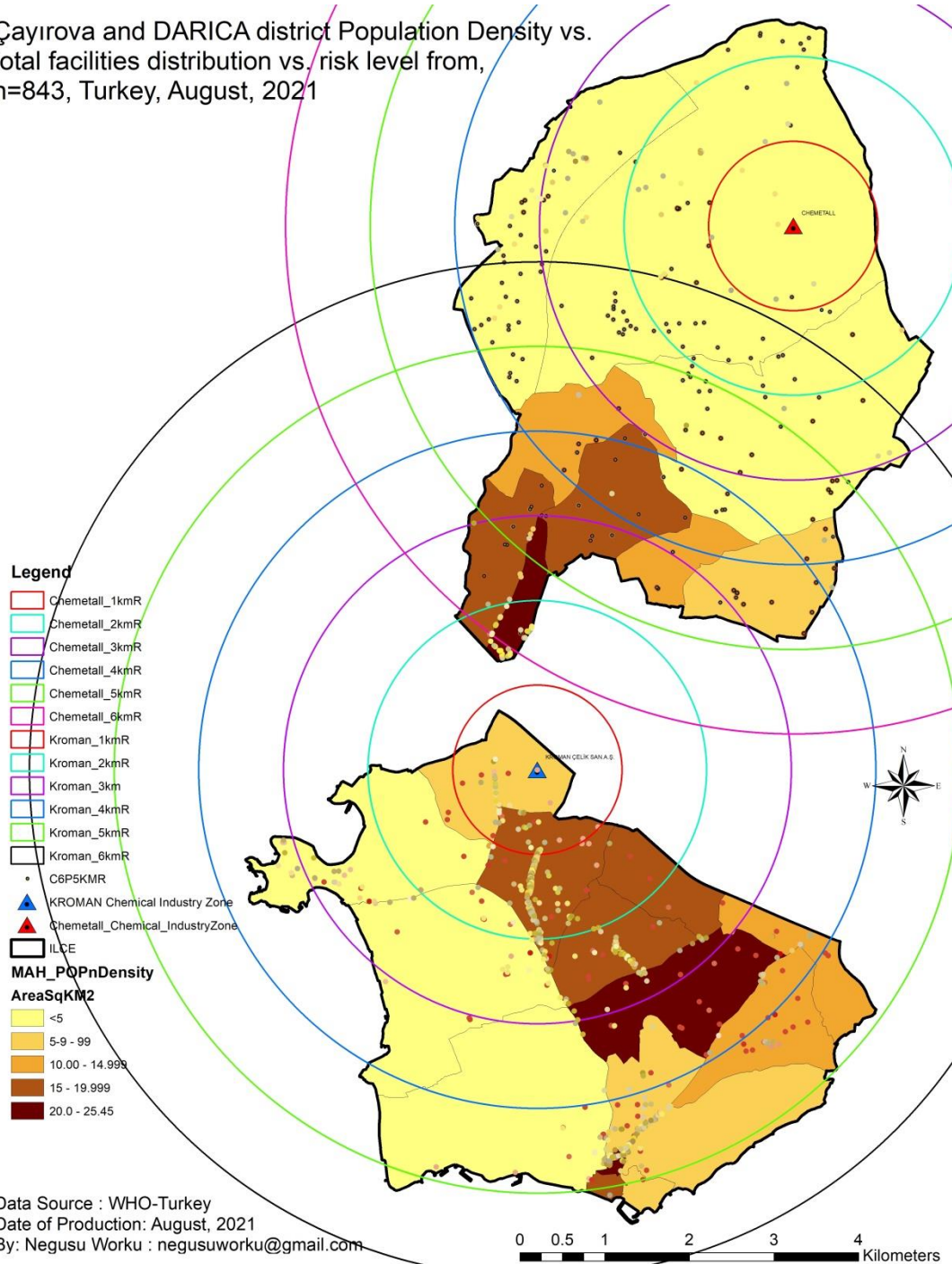
Of which n=205 of them were found to be within 1kmR. The majority of the services are high risk.

The majority of n=199 are unclassified buildings, n=5 are Factory/Industry

Cayırova and Darıca Districts Distribution of total facilities by risk level by different kilometer radius, n=843, Turkey, August, 2021

KMRadius	District	Number of Buildings	Risk Level	List of Facilities
1kmR	Cayırova	5	High Risk	E:\n843dbf\C1KMr843.xlsx
2kmR	Cayırova	40	Medium risk	E:\n843dbf\C2KMr843.xlsx
3kmR	Cayırova	95	Low risk	E:\n843dbf\C3KMr843.xlsx
4kmR	Cayırova	149	Low risk	E:\n843dbf\C4KMr843.xlsx
5kmR	Cayırova	207	Low risk	E:\n843dbf\C5KMr843.xlsx
6kmR	Cayırova	222	Low risk	E:\n843dbf\C6KMr843.xlsx
>6kmR	Cayırova	238	Low risk	E:\n843dbf\C6P5KMR.xlsx
KMRadius	District	Number of Buildings	Risk Level	
1kmR	Darica	49	High Risk	E:\n843dbf\k1KMr843.xlsx
2kmR	Darica	257	Medium risk	E:\n843dbf\k2KMr843.xlsx
3kmR	Darica	373	Low risk	E:\n843dbf\k3KMr843.xlsx
4kmR	Darica	445	Low risk	E:\n843dbf\k4KMr843.xlsx
5kmR	Darica	678	Low risk	E:\n843dbf\k5KMr843a.xlsx
6kmR	Darica	750	Low risk	E:\n843dbf\k6KMr843.xlsx

Çayırova and DARICA district Population Density vs. total facilities distribution vs. risk level from, n=843, Turkey, August, 2021

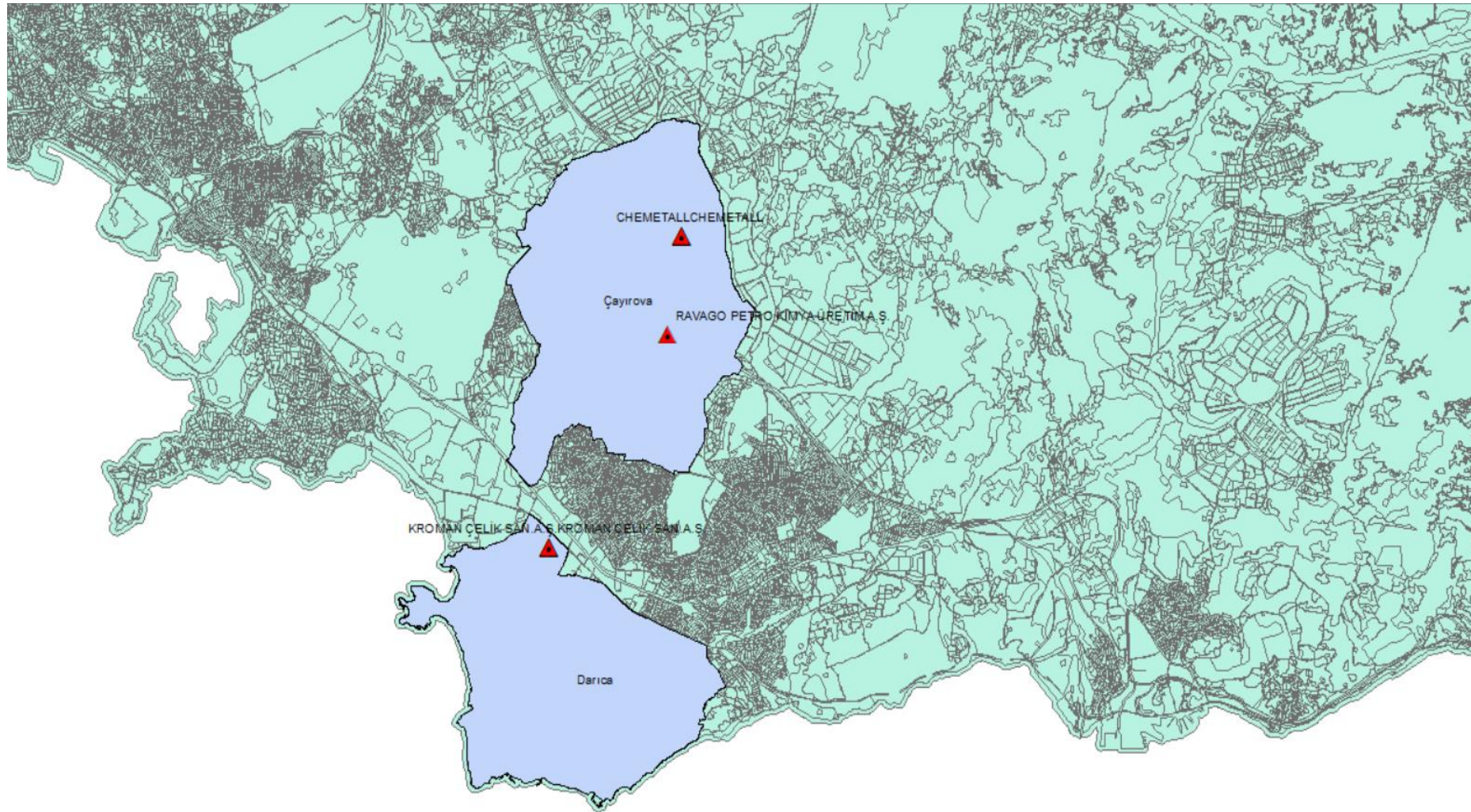


From the total **n=843** point of interest

At Darica district, from the Kroman Chemical industry **n=49 facilities** are located within 1kmR including : Schools, Industry, Health facilities, Shops, Café, Bakery and Hotels

From Cayirova district, **n=5** facilities are located of which all n=3 of them are industry n=1 Fuel gas station and n=1 car parts,

3. Converting the CORINE from Raster to Polygon. This helps to do the analysis very easy. To Clip CORINE.TIF converted to CORINE.SHP but the back end of reading such as : grid code of :



1. The above feature is converted from Raster to Polygon (from TIF format to Shape file)
2. Clipped CORINE based on the territory of **Çayirova and Darica District**,
3. But the questions is : which Admin2 of CAYIROVA and DARICA is correct? From WHO/MOH, from UN-OCHA-HDX or any other source? For the sake of continuing the Exercise, I worked with the file source from WHO/MOH.

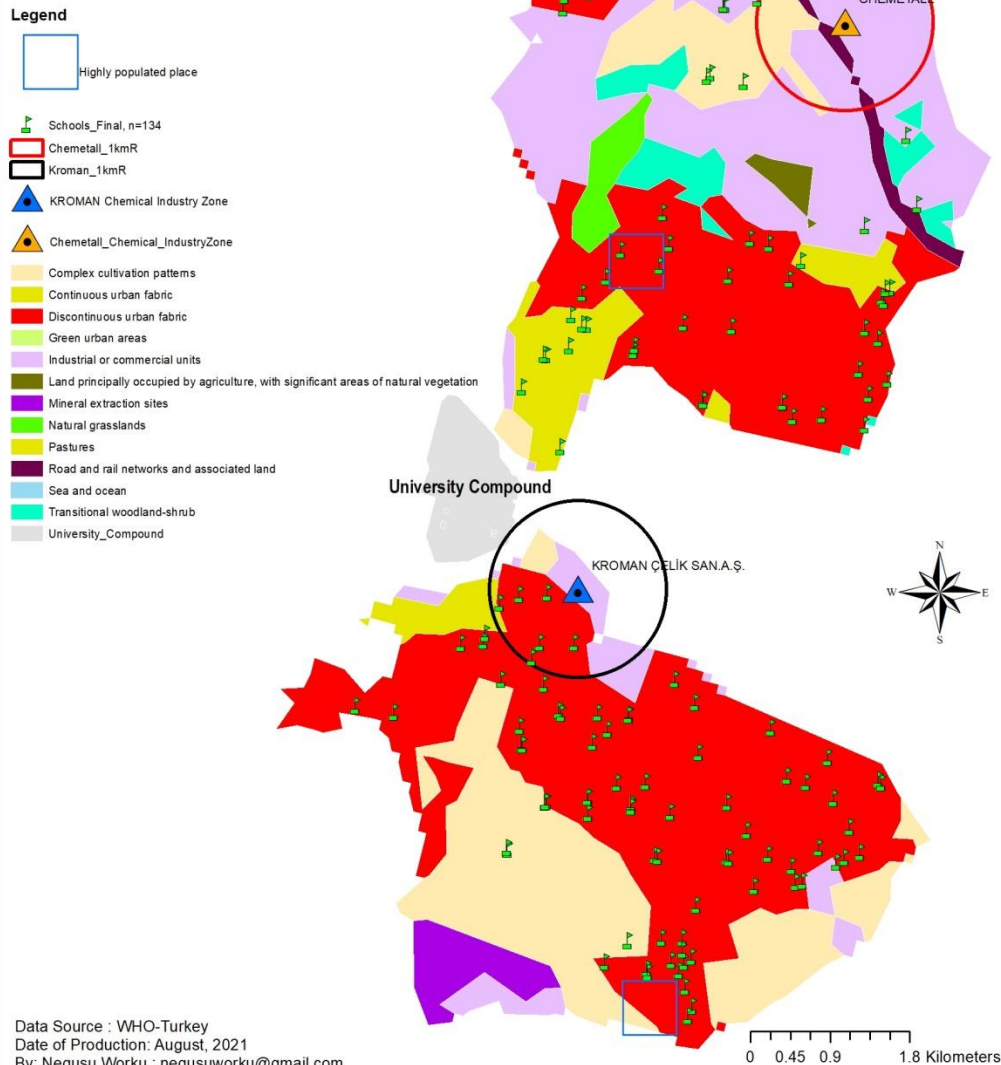
The CORINE features distribution in Çayırova and Darıca District

Classification of CORINE for CAYIROVA and DARICA	number of observation	%
Discontinuous dense urban fabric (S.L. : 50% - 80%)	1651	39.1
Continuous urban fabric (S.L. : > 80%)	1047	24.8
Industrial, commercial, public, military and private units	404	9.6
Discontinuous medium density urban fabric (S.L. : 30% - 50%)	241	5.7
Land without current use	193	4.6
Pastures	170	4.0
Discontinuous low density urban fabric (S.L. : 10% - 30%)	110	2.6
Discontinuous very low density urban fabric (S.L. : < 10%)	102	2.4
Arable land (annual crops)	93	2.2
Green urban areas	74	1.8
Herbaceous vegetation associations (natural grassland, moors...)	49	1.2
Mineral extraction and dump sites	32	0.8
Construction sites	24	0.6
Sports and leisure facilities	21	0.5
Railways and associated land	4	0.1
Fast transit roads and associated land	3	0.1
Forests	2	0.0
Isolated structures	1	0.0
Open spaces with little or no vegetation (beaches, dunes, bare rocks, glaciers)	1	0.0
Other roads and associated land	1	0.0
Port areas	1	0.0
Water	1	0.0
Total	4225	100.0

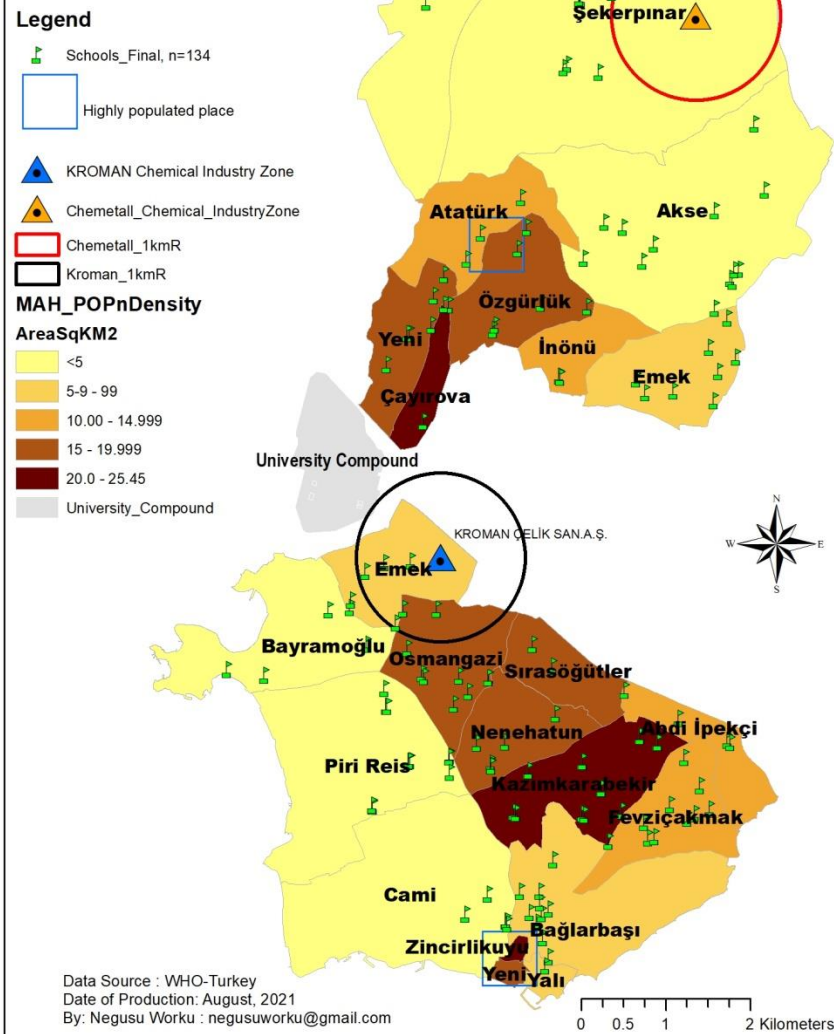
The CORINE features Distribution in Çayırova and Darıca

- **64%** of the land of Çayırova and Darıca District is covered by discontinuous Urban fabric and Continuous Urban fabric,
- **Nearly 10%** of the land of Çayırova and Darıca District covered by Industrial, Commercial, Public and military and PVT Units, (Please see the distribution in the converted map from Raster to Polygon),

Çayırova and DARICA district vs CORINE land distribution
VS School Facilities vs Chemical Industry,Turkey, Augst, 2021



Çayırova and DARICA district Population Density vs School
Facilities vs Chemical Industry,Turkey, August, 2021,



CORINE land vs. Schools Correlation Findings

- **Chemetall Industry Zone from Cayirova District**
 - CORINE Covers **within 1kmR**,
 - **70%** industry,
 - **20%** Complex Cultivation and
 - **10%** road,
 - Low population density
 - Less risk of facilities from the point of Chemetall industry zone,
 - Only n=1 School available within 1 kmR,
- **Kroman Industry from Darıca District**
 - **40%** Sea
 - **35%** Discontinues urban fabric,
 - **10%** Complex Cultivation Pattern,
 - High Population density,
 - Within 1kmR risk there are n=6 schools at risk,
 - University Compound is at risk within 1kmR,

Çayırova and DARICA district vs CORINE land distribution
VS Health Facilities vs Chemical Industry,Turkey, August, 2021

Legend

HF_s_n44_Final

Health Facility Category

+ Health Center

[H] Hospital

+ Pharmacy

[] Highly populated place

[] Chemetall_1kmR

[] Kroman_1kmR

[] KROMAN Chemical Industry Zone

[] Chemetall_Chemical_IndustryZone

[] Complex cultivation patterns

[] Continuous urban fabric

[] Discontinuous urban fabric

[] Green urban areas

[] Industrial or commercial units

[] Land principally occupied by agriculture, with significant areas of natural vegetation

[] Mineral extraction sites

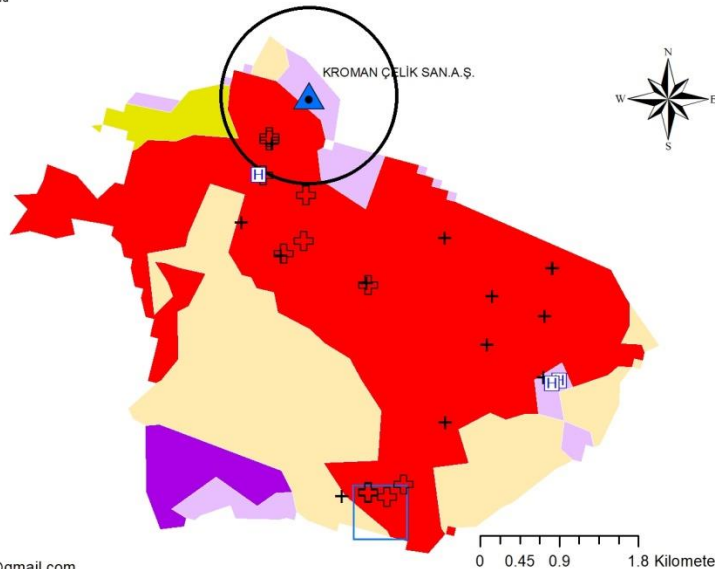
[] Natural grasslands

[] Pastures

[] Road and rail networks and associated land

[] Sea and ocean

[] Transitional woodland-shrub



Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com

Çayırova and DARICA district Population Density vs Health
Facilities vs Chemical Industry,Turkey, August, 2021,

Legend

HF_s_n44_Final

Health Facility Category

+ Health Center

[H] Hospital

+ Pharmacy

[] Highly populated place

[] KROMAN Chemical Industry Zone

[] Chemetall_Chemical_IndustryZone

[] Chemetall_1kmR

[] Kroman_1kmR

MAH_POPnDensity

AreaSqKM2

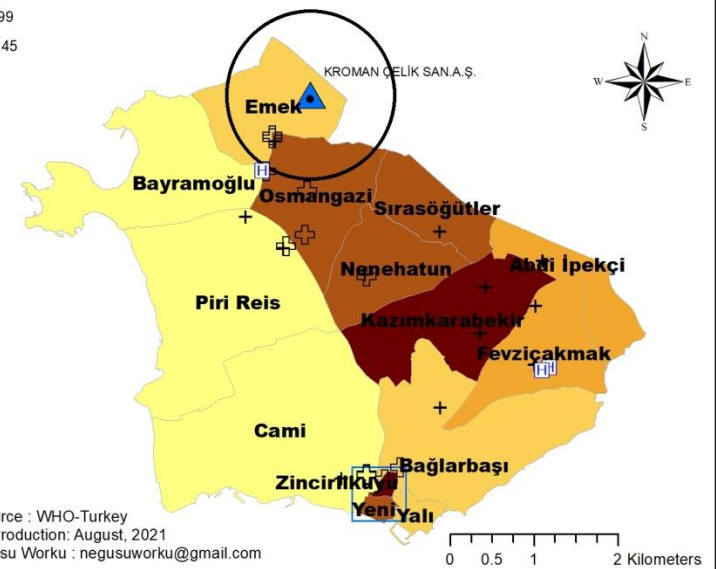
[] <5

[] 5-9 - 99

[] 10.00 - 14.999

[] 15 - 19.999

[] 20.0 - 25.45

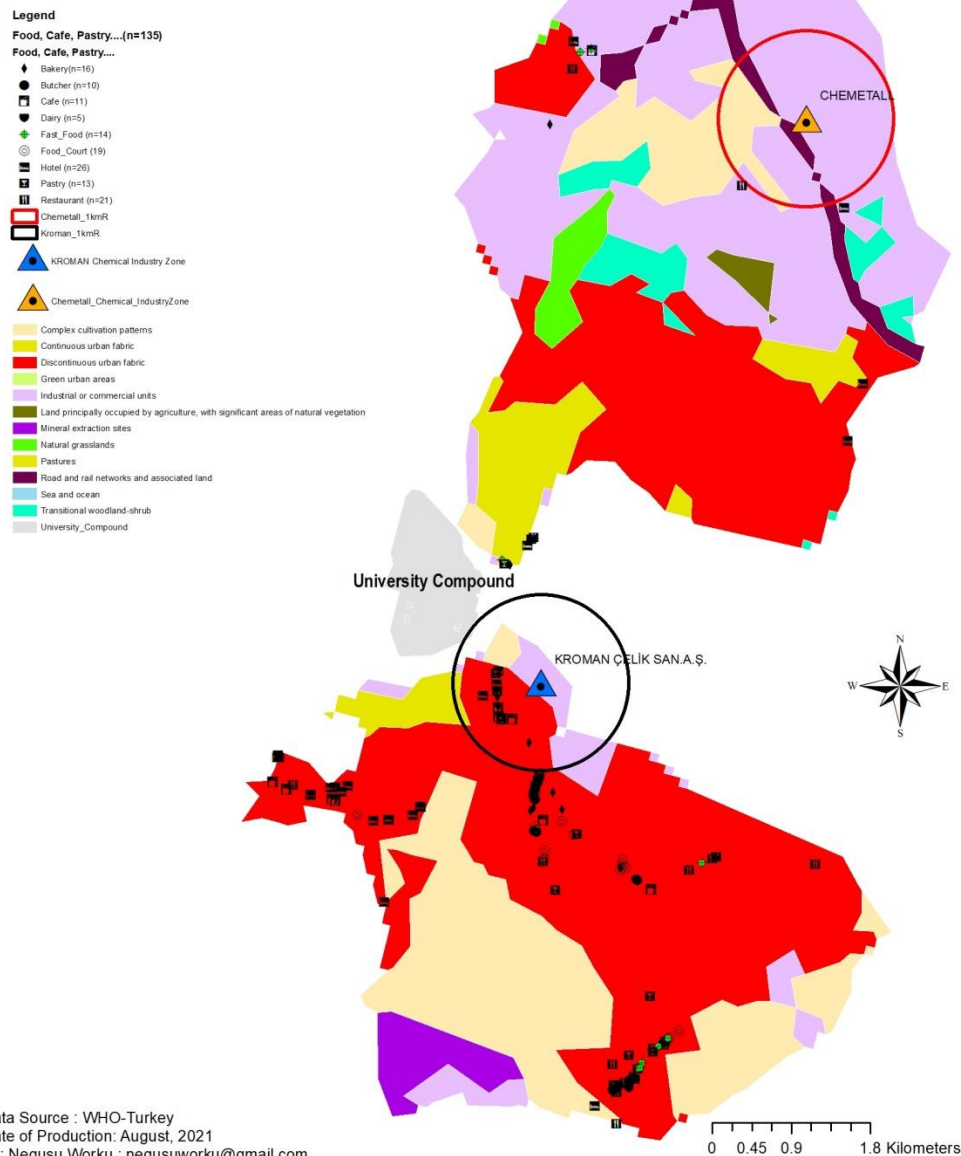


Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com

CORINE vs. Health Facilities

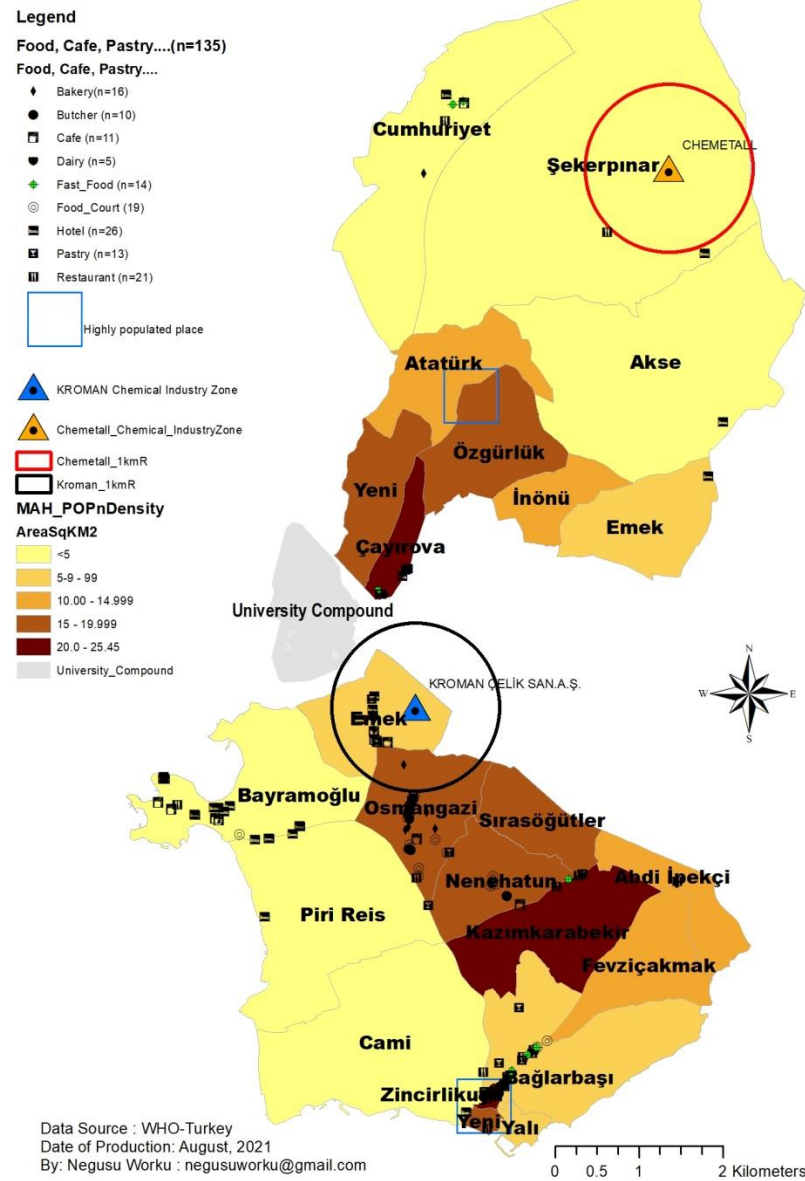
- Cayirova District
 - No facility within 1kmR,
 - No HFs in the North east of Cayirova,
 - All the HFs concentrated in the South part of Cayirova, where population density is high,
 - High Population place is in the South of Cayirova,
 - Health facility is located in 3km from the Chemetall Chemical Industry,
- Kroman District,
 - Within 1 kmR,
 - 1HC
 - 3 Pharmacy
 - 1 Hospital available,
 - Popn density is high,
 - HFs available in the south eastern part of Kroman, where population density is high and high populated place,
 - Railway route is also available in the northern tip of Darica and located 1kmR from the Kroman chemical Industry Zone,

Çayırova and DARICA district vs CORINE land distribution
VS Food, Cafe, Pastry, Fast Food and etcvs
Chemical Industry,Turkey, August, 2021



Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com

Çayırova and DARICA district Population Density vs
Food, Cafe, Pastry, Fast Food and etc vs
Chemical Industry,Turkey, August, 2021,

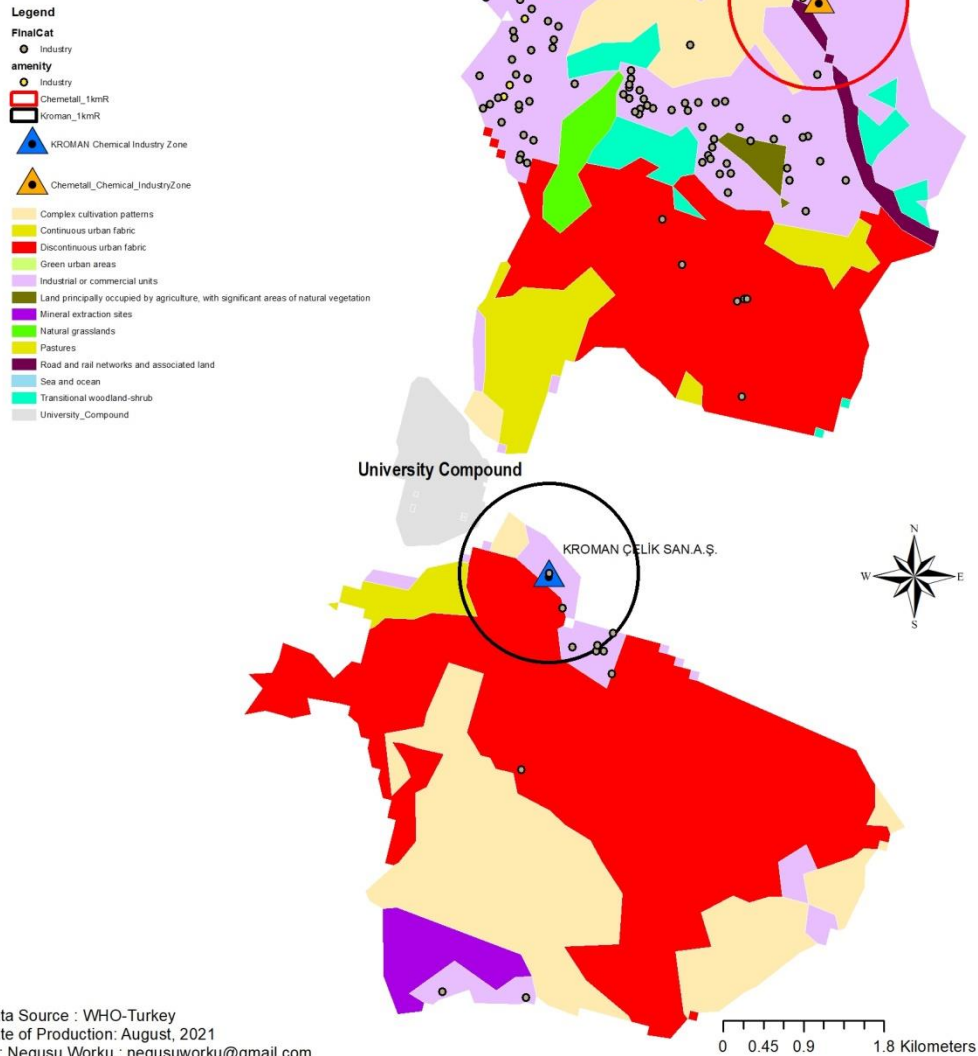


Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com

CORINE vs. Food Industry

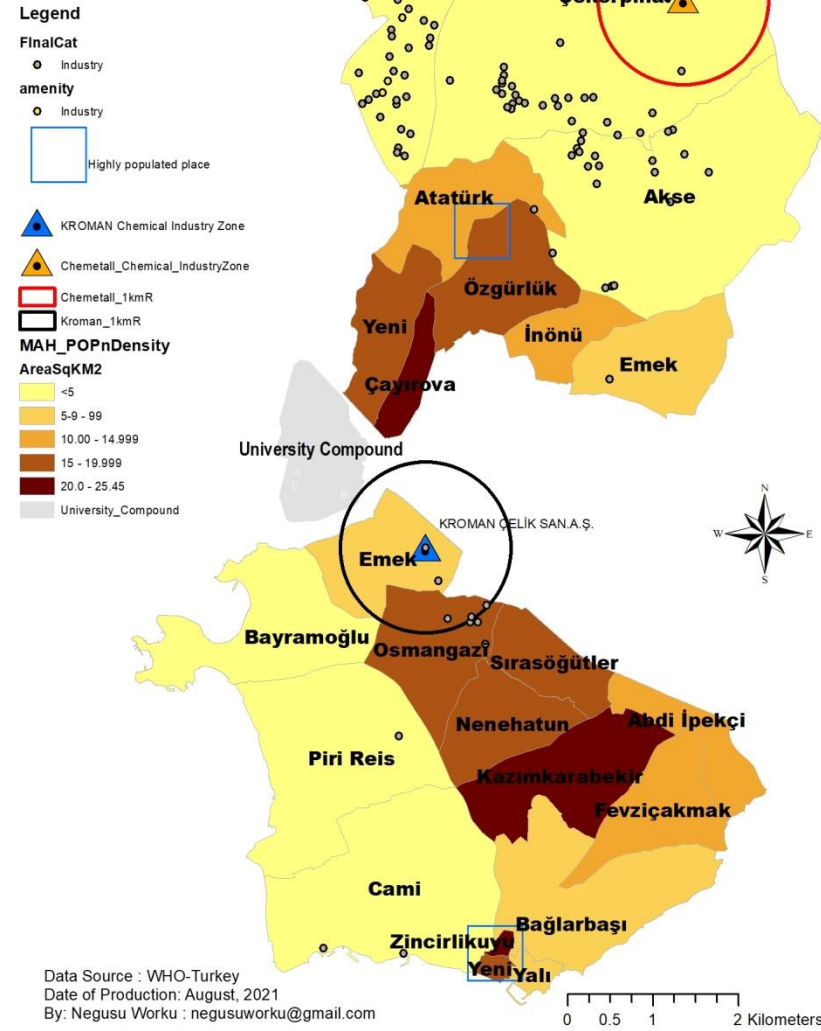
- Within 1kmR from the Kroman Chemical Industry:
 - 14 food industry available including:
 - n=4 Bakery
 - n=4 Café
 - n=3 Hotels
 - n=2 Pastry
 - n=1 Restaurant
 - No food industry within 1 kmR from Chemetall Chemical Industry, Cayirova district,

Çayırova and DARICA district vs CORINE land distribution
VS Industry Zone (n=110) vs Chemical Industry,Turkey, Augst, 2021



Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com

Çayırova and DARICA district Population Density vs
Industry vs Chemical Industry (n=110),Turkey, August, 2021,

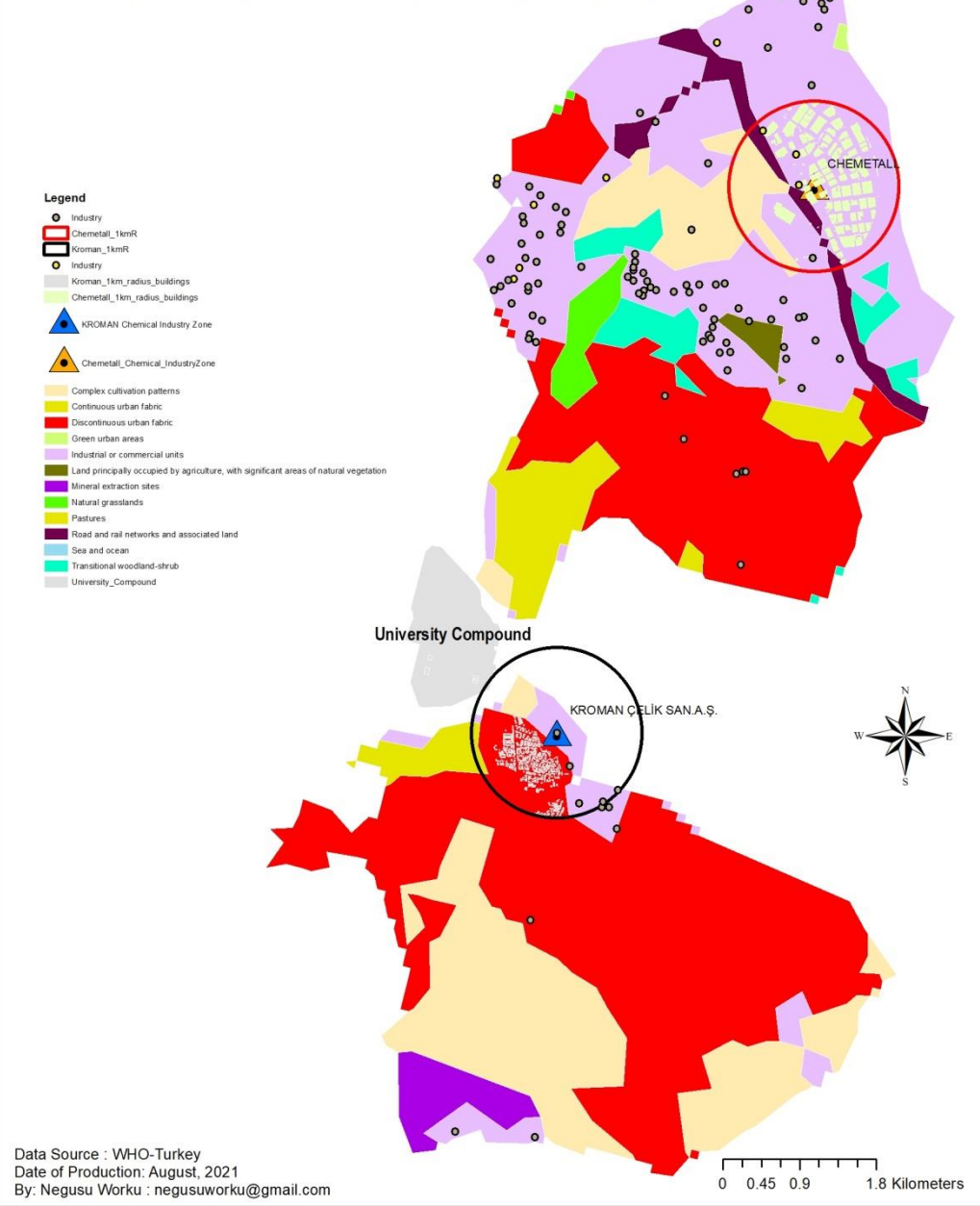


Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com

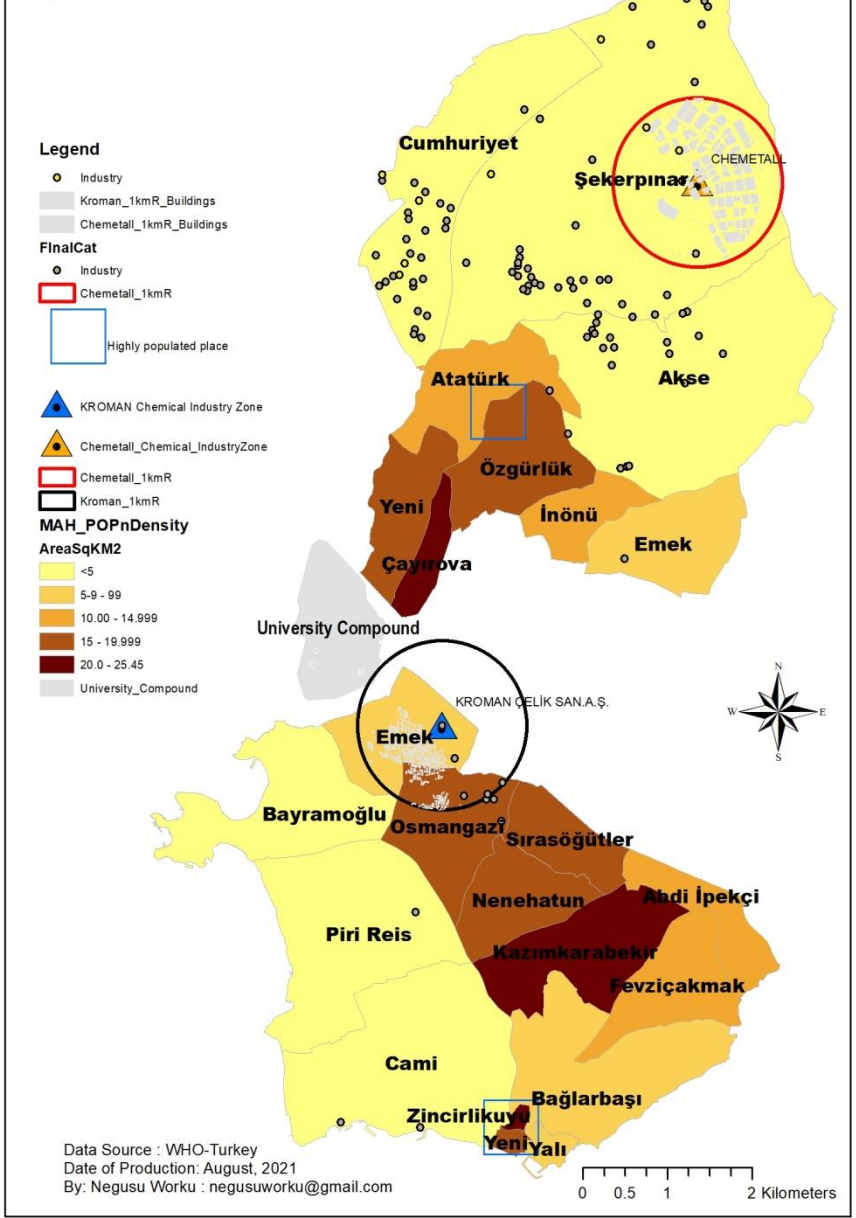
CORINE vs. Industries

- From n=110 industry zone, n=7 (6%) located out of the Industry zone (located in the community/in the urban fabric zone),
- n=103(94%) of the industries from the two districts located in the industry zone,
- From the Chemetall chemical industry, from Cayirova district, n=4 industries located in 1kmradius in the industry zone and
- From Kroman Chemical industry, n=7 industries located within 1km radius where correctly fall in the industry zone (within 1km radius)

Çayırova and DARICA district vs CORINE land distribution
VS buildings within 1kmR vs Chemical Industry,Turkey, Augst, 2021



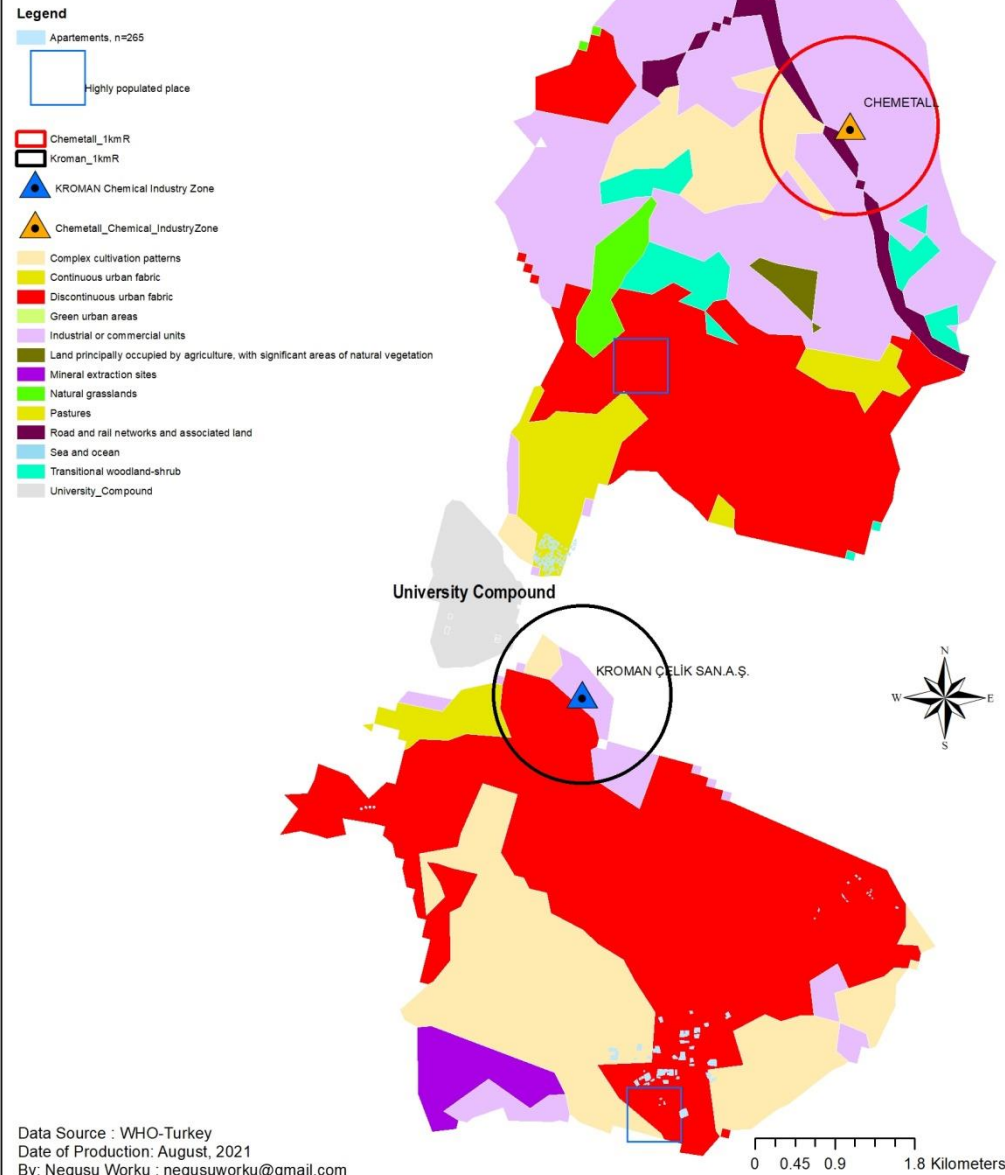
Çayırova and DARICA district Population Density vs buildings within 1kmR vs Chemical Industry,Turkey, August, 2021,



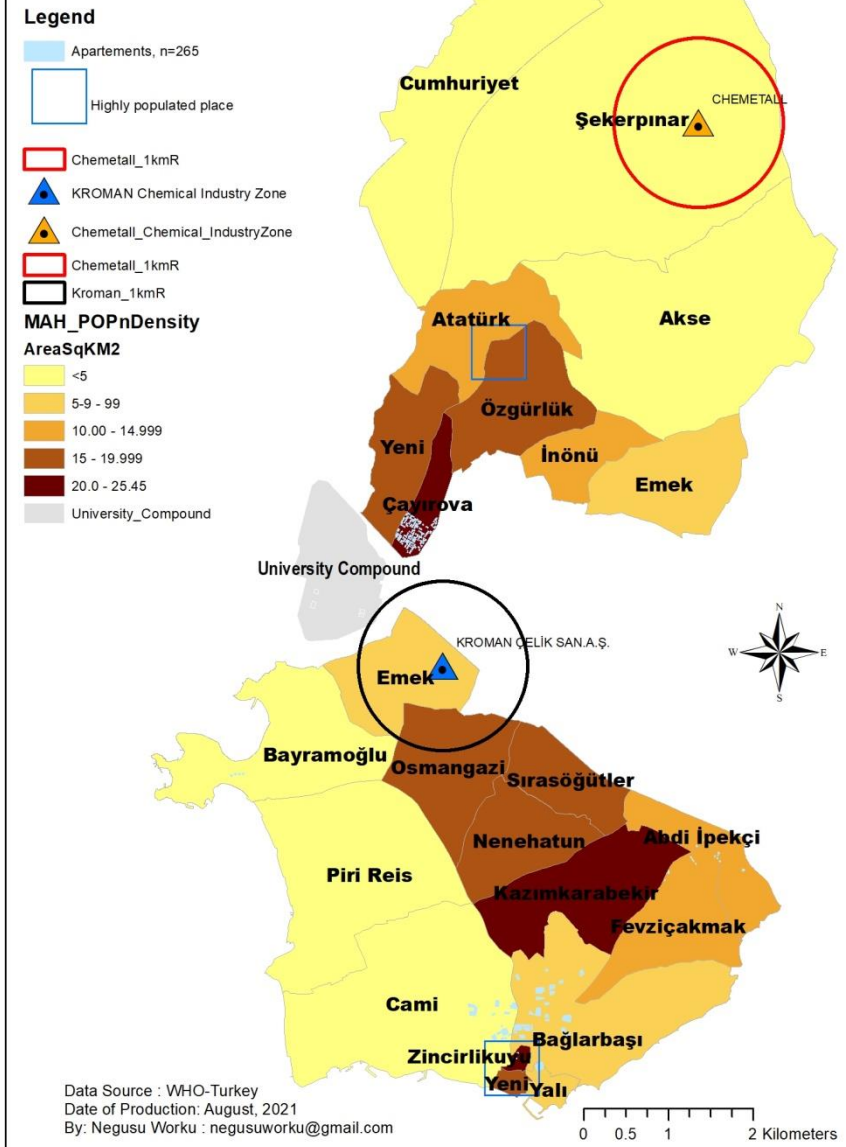
CORINE vs. Buildings

- n=205 buildings are available in the Chemetall chemical industry, within 1km radius, Cayirova districts,
- n=708 buildings are available in the Kroman Chemical industry within 1kmR from Darica district

Çayırova and DARICA district vs CORINE land distribution
VS apartment building (n=265) within 1kmR vs Chemical Industry,
Turkey, Augst, 2021



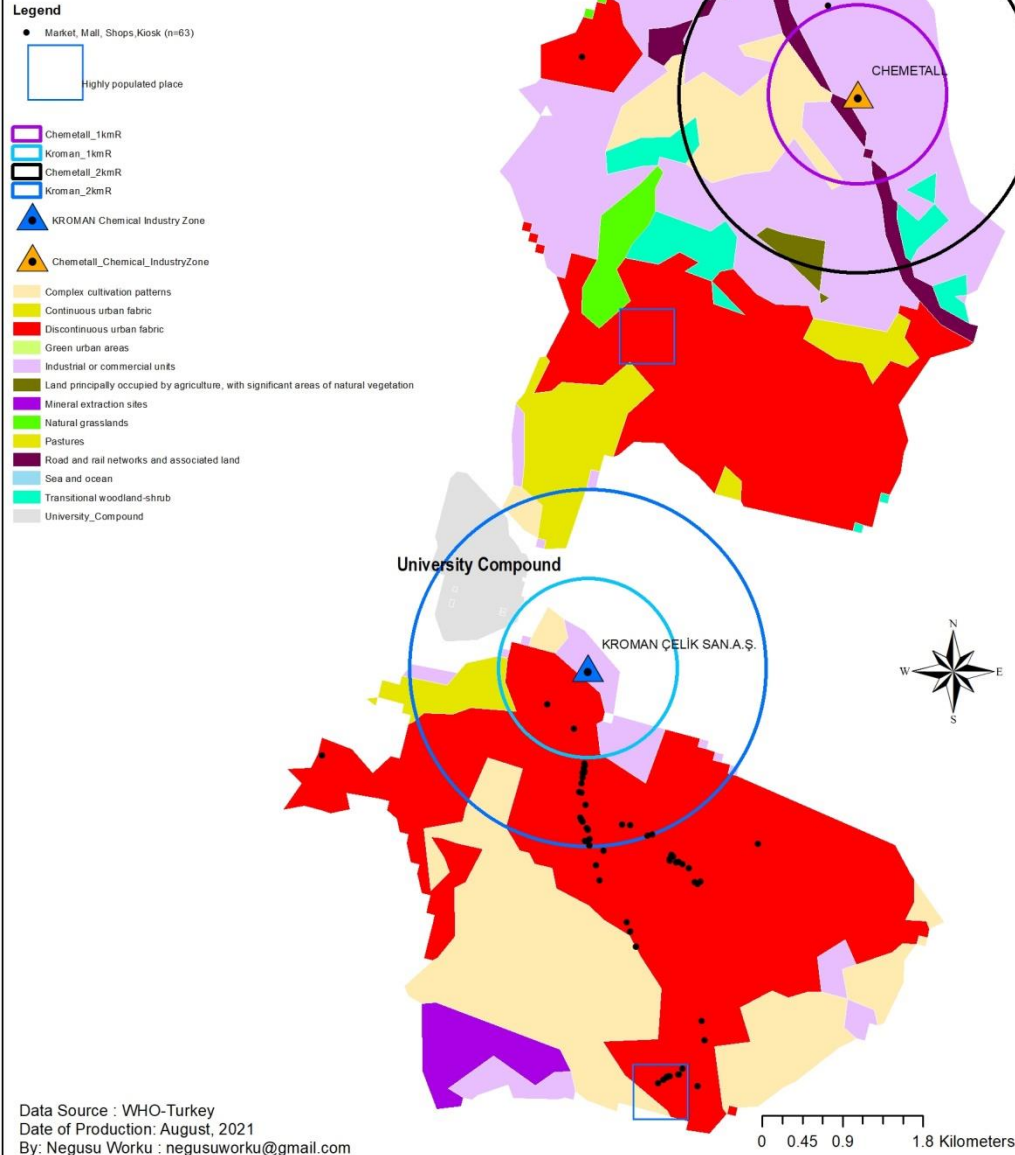
Çayırova and DARICA district Population Density vs apartments
buildings within 1kmR vs Chemical Industry,Turkey,
August, 2021,



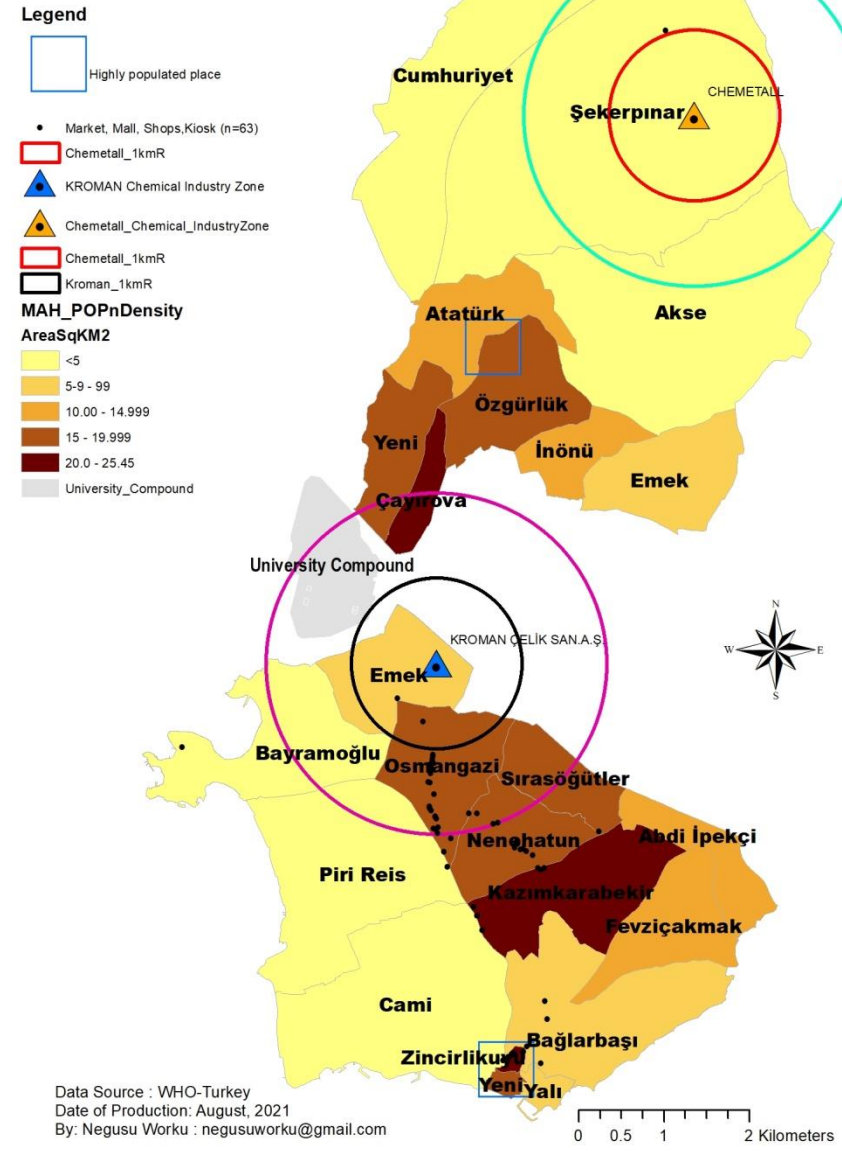
CORINE vs. Apartments

- Cayirova, apartments are located the southern part and no chemical industry
- Darica, apartments are located in the southern part of, no chemical industry and population density is high,

Çayırova and DARICA district vs CORINE land distribution
VS market place, kiosk, mall and shops (n=63) within
1 & 2 kmR vs Chemical Industry,
Turkey, August, 2021



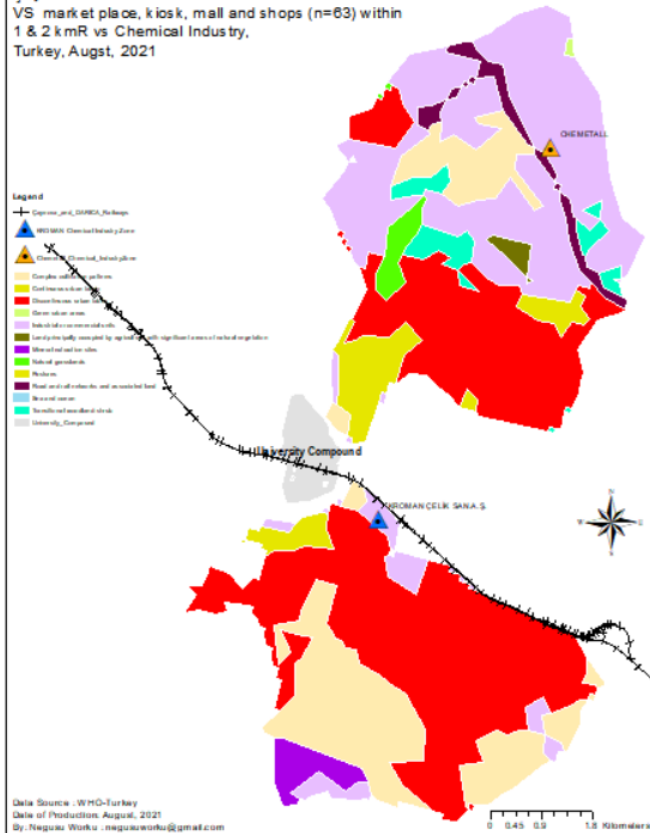
Çayırova and DARICA district Population Density vs market place,
kiosk, mall and shops (n=63) within
1 & 2 kmR vs Chemical Industry, Turkey,
August, 2021,



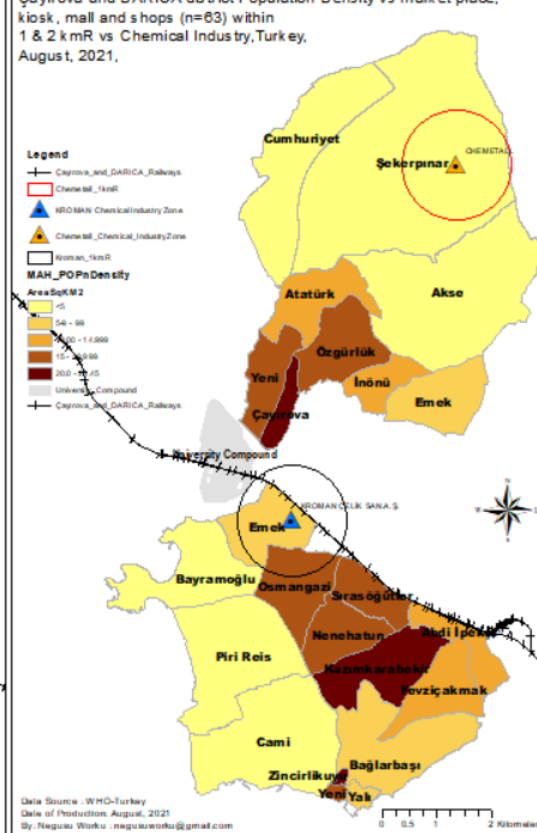
CORINE vs. Malls, Shops, Kiosk and Market place

- Kroman Chemical Industry Zone,
 - 1kmR = 2 shops
 - 2kmR = 26 facilities including :
 - n=4 malls
 - n=22 shops
- Cayirova Chemical Industry Zone,
 - 1kmR = No facilities,
 - 2kmR = only 1 shops

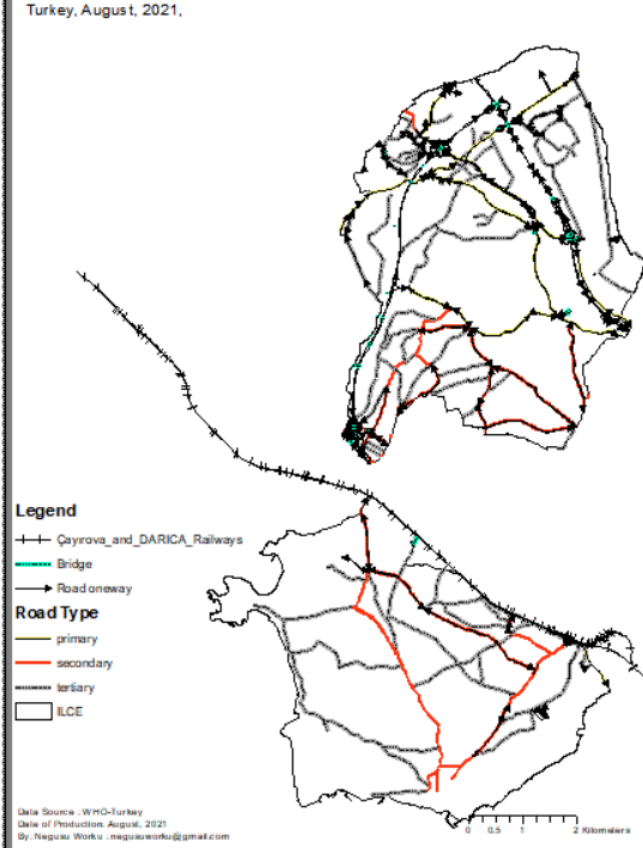
Çayirova and DARICA district vs CORINE land distribution
VS market place, kiosk, mall and shops (n=63) within
1 & 2 kmR vs Chemical Industry,
Turkey, August, 2021



Çayirova and DARICA district Population Density vs market place,
kiosk, mall and shops (n=63) within
1 & 2 kmR vs Chemical Industry, Turkey,
August, 2021.



Çayirova and DARICA district Population Density vs Road, Rail and Bridge Type,
Turkey, August, 2021.



Base map of :

1. Land distribution
2. Population density and
3. Road type

Types of Building distribution, n=5770, Çayirova and DARICA districts, July, 2021, Turkey

Buildings	Frequency	Percent
Yes	5120	88.7
Apartments	265	4.6
House	230	4.0
Industrial	37	.6
Roof	30	.5
School	26	.5
Greenhouse	18	.3
Residential	8	.1
Shed	7	.1
Commercial	6	.1
Farm	5	.1
Warehouse	4	.1
Dormitory	2	.0
Garage	2	.0
hospital	2	.0
Hotel	2	.0
Civic	1	.0
Factory	1	.0
Guard_house	1	.0
Kindergarten	1	.0
Office	1	.0
Public	1	.0
Total	5770	100.0

From the Kroman Chemical industry, buldings observed 1km radius, n=708, Darica

Buildings	Number	%
Commercial	1	0.1
Hotel	1	0.1
Roof	1	0.1
School	4	0.6
House	52	7.3
Yes	649	91.7
Total	708	100.0

From the Chemetall Chemical industry, buldings observed 1km radius, n=205, Çayirova

Buildings	Frequency	Percent
Factory	1	0.5
industrial	4	2.0
Roof	1	0.5
Yes	199	97.1
Total	205	100.0

Summary of Facilities based on km radius, Çayırova district, Chemetall Chemical Industry, July, 2021, Turkey

Distribution of facilities within 1m radius from the point of Chemetall chemical industry, n=205

Buildings	# of observations	%
Factory	1	0.5
Industrial	4	2.0
Roof	1	0.5
Yes	199	97.1
Total	205	

Frequency of facilities within 2km radius from the point of Chemetall chemical industry

Buildings	number of observations	%
Factory	1	0.3
Industrial	11	3.2
Pubic	1	0.3
Roof	2	0.6
Yes	328	95.6
Total	343	100.0

Frequency of facilities within 3km radius from the point of Chemetall chemical industry

Buildings	number of observations	%
Factory	1	0.2
House	1	0.2
Industrial	11	2.7
Public	1	0.2
Roof	7	1.7
School	2	0.5
Yes	381	94.3
Total	404	100.0

List of facilities within 4km radius from the point of Chemetall chemical industry

Building	Number of observations	%
Civic	1	0.2
Commercial	1	0.2
Factory	1	0.2
House	1	0.2
Industrial	11	2.5
Public	1	0.2
Roof	14	3.1
School	6	1.3
Warehouse	4	0.9
Yes	408	91.1
Total	448	100.0

Frequency of facilities within 5km radius from the point of Chemetall chemical industry

Buildings	# of observaion	%
Civic	1	0.2
Commercial	1	0.2
Factory	1	0.2
Hospital	1	0.2
House	1	0.2
Idustrial	13	2.4
Public	1	0.2
Roof	17	3.1
School	11	2.0
Warehouse	4	0.7
Yes	490	90.6
Total	541	100.0

Frequency of facilities within 6km radius from the point of Chemetall chemical industry

Buildings	Frequency	Percent
civic	1	0.1
commercial	1	0.1
Factory	1	0.1
hospital	1	0.1
house	2	0.2
industrial	13	1.3
public	1	0.1
roof	17	1.7
school	11	1.1
warehouse	4	0.4
yes	941	94.8
Total	993	100.0

Frequency of facilities within 6.5km radius from the point of Chemetall chemical industry

Building	Frequency	Percent
Apartments	191	11.4
Civic	1	0.1
Commercial	1	0.1
Factory	1	0.1
Hospital	1	0.1
House	57	3.4
Industrial	13	0.8
Public	1	0.1
Roof	22	1.3
School	12	0.7
Warehouse	4	0.2
Yes	1365	81.8
Total	1669	100.0

Buildings	District		Total
	Cayiravo	Darica	
Yes	1365	3755	5120
Apartments	191	74	265
House	57	173	230
Roof	22	8	30
Industrial	13	24	37
School	12	14	26
Warehouse	4	0	4
Civic	1	0	1
Commercial	1	5	6
Factory	1	0	1
Hospital	1	1	2
Public	1	0	1
Dormitory	0	2	2
Farm	0	5	5
Garage	0	2	2
Greenhouse	0	18	18
Guard_house	0	1	1
Hotel	0	2	2
Kindergarten	0	1	1
Office	0	1	1
Residential	0	8	8
Shed	0	7	7
Total	1669	4101	5770

Summary of Building facilities (n=5770) with level of KM radius, Cayirova and Darİca, Chemetall and Koroman Chemical industry, August, 2021, Turkey

	Chemical		1kmR	2kmR	3kmR	4kmR	5kmR	6kmR
District	industry Zone	Types of Building	(n=205)	(n=343)	(n=404)	(n=448)	(n=541)	(n=1463)
Cayirova	Chemettal Cheical Industry	Apartments	0	0	0	0	0	72
Cayirova		Civic	0	0	0	1	1	1
Cayirova		Commercial	0	0	0	1	1	1
Cayirova		Factory	1	1	1	1	1	1
Cayirova		Hospital	0	0	0	0	1	1
Cayirova		House	0	0	1	1	1	17
Cayirova		Industry	4	11	11	11	13	13
Cayirova		Public	0	1	0	1	1	1
Cayirova		Roof	1	2	7	14	17	21
Cayirova		School	0	0	2	6	11	12
Cayirova		Warehouse	0	0	0	4	4	4
Cayirova		Yes (Uncatagorized Buld)	199	328	381	408	490	1319
Total			205	343	404	448	541	1463
DariCa	Koroma Chemical Industry Zone	Apartments	0	0	191	197	264	265
DariCa		Commercial	1	0	1	1	5	5
DariCa		Hotel	1	0	1	2	2	2
DariCa		Hospital	0	0	1	1	2	2
DariCa		House	52	0	107	109	163	229
DariCa		Industry	0	11	0	0	24	24
DariCa		Public	0	1	0	0	0	0
DariCa		Roof	1	2	9	9	13	17
DariCa		School	4	0	8	9	16	18
DariCa		Dormitory	0	0	0	1	2	2
DariCa		Residential	0	0	0	3	8	8
DariCa		Farm	0	0	0	0	0	5
DariCa		Garage	0	0	0	0	0	2
DariCa		Greenhouse	0	0	0	0	0	18
DariCa		Kindergarten	0	0	0	0	0	1
DariCa		Yes (Uncatagorized Bul	649	328	2965	3662	4629	5293
District	Total		708	343	3284	3662	4629	5293

Summary of Point of Interest facilities (n=260) with level of KM radius, Cayirova, Chemetall and Koroman Chemical industry, August, 2021, Turkey

District	Chemical industry Zone	Types of Service	1kmR	2kmR	3kmR	4kmR	5kmR	6kmR	6.5kmR
Cayirova	Chemical industry Zone	Airport		1	1	1	1	1	1
Cayirova	Chemical industry Zone	Bakery				1	1	2	2
Cayirova	Chemical industry Zone	Bank			1	1	1	5	5
Cayirova	Chemical industry Zone	Bench				2			
Cayirova	Chemical industry Zone	Bus station			1	1	1	2	2
Cayirova	Chemical industry Zone	Café				1	1	1	1
Cayirova	Chemical industry Zone	Car Part	1	3	3	3	5	6	6
Cayirova	Chemical industry Zone	Car rental		1	1	1	1	3	3
Cayirova	Chemical industry Zone	Charging station		2	2	2	2	2	2
Cayirova	Chemical industry Zone	Dormitory				1	3	3	3
Cayirova	Chemical industry Zone	Fast Food				2	2	2	2
Cayirova	Chemical industry Zone	Fuel	1	3	7	12	14	14	14
Cayirova	Chemical industry Zone	Health Center			1	3	9	11	11
Cayirova	Chemical industry Zone	Hospital				1	1	1	1
Cayirova	Chemical industry Zone	Hotel		1	1	5	5	6	6
Cayirova	Chemical industry Zone	Industry	3	19	57	69	96	98	98
Cayirova	Chemical industry Zone	Marketplace					1	1	1
Cayirova	Chemical industry Zone	Pharmacy				1	1	2	2
Cayirova	Chemical industry Zone	Post Office						1	1
Cayirova	Chemical industry Zone	Restaurant		1	1	2	2	4	4
Cayirova	Chemical industry Zone	School		7	16	32	49	57	57
Cayirova	Chemical industry Zone	Store		1	1	1	1	1	1
Cayirova	Chemical industry Zone	Supermarket						3	3
Cayirova	Chemical industry Zone	Town hall			1	2	2	2	2
Cayirova	Chemical industry Zone	Tyres				1	1	1	1

**Summary of Point of Interest facilities (n=583) with level of KM radius, Darica, Koroman
Chemical industry, August, 2021, Turkey**

District	Chemical industry Zone	Types of Service	1kmR	2kmR	3kmR	4kmR	5kmR	6kmR	6.5kmR
Cayırova	Chemical industry Zone	Airport		1	1	1	1	1	1
Cayırova	Chemical industry Zone	Bakery				1	1	2	2
Cayırova	Chemical industry Zone	Bank			1	1	1	5	5
Cayırova	Chemical industry Zone	Bench				2			
Cayırova	Chemical industry Zone	Bus station			1	1	1	2	2
Cayırova	Chemical industry Zone	Café				1	1	1	1
Cayırova	Chemical industry Zone	Car Part	1	3	3	3	5	6	6
Cayırova	Chemical industry Zone	Car rental		1	1	1	1	3	3
Cayırova	Chemical industry Zone	Charging station		2	2	2	2	2	2
Cayırova	Chemical industry Zone	Dormitory				1	3	3	3
Cayırova	Chemical industry Zone	Fast Food				2	2	2	2
Cayırova	Chemical industry Zone	Fuel	1	3	7	12	14	14	14
Cayırova	Chemical industry Zone	Health Center			1	3	9	11	11
Cayırova	Chemical industry Zone	Hospital				1	1	1	1
Cayırova	Chemical industry Zone	Hotel		1	1	5	5	6	6
Cayırova	Chemical industry Zone	Industry	3	19	57	69	96	98	98
Cayırova	Chemical industry Zone	Marketplace					1	1	1
Cayırova	Chemical industry Zone	Pharmacy				1	1	2	2
Cayırova	Chemical industry Zone	Post Office						1	1
Cayırova	Chemical industry Zone	Restaurant		1	1	2	2	4	4
Cayırova	Chemical industry Zone	School		7	16	32	49	57	57
Cayırova	Chemical industry Zone	Store		1	1	1	1	1	1
Cayırova	Chemical industry Zone	Supermarket						3	3
Cayırova	Chemical industry Zone	Town hall			1	2	2	2	2
Cayırova	Chemical industry Zone	Tyres				1	1	1	1

Major Findings

- High population density in the Southern part of Çayırova district,
- High population density in the north eastern part of Darica,
- From the Chemetall Chemical industrial zone, within 1km radius, there is no facilities (Schools and HFs),
- There are two schools and health facilities at the Darica district within one kilometer radius,
- 24% of Health facilities (HFs) located at Çayırova district. These facilities are located beyond 3km radius from the point of Chemetall industry zone,
- 76% of Health facilities located at Darica district. These facilities are located within One, two and three kilo meter radius and the population density is high,
- 14% of schools located at Çayırova district,
- 86% schools are located at Darica district, the population density is high at Darica,
- Schools and Health facilities are located within One kilometer radius from the Kroman Chemical industry Zone along the CORINE land distribution of Urban fabric, Industrial or Commercial unit and Complex cultivation pattern,
- At Çayırova district, population density is less and below 5 person per square kilometer and the health facilities and Schools are far beyond 3km radius and the CORINE land cover is Industrial Commercial Units, Complex Cultivation pattern and road. But the health facilities with the Çayırova district located at the South edge, more closer to he Darica district. These facilities and Schools are near to 2km radius from the point of Kroman Chemical Industry. Moreover, population density is high at the Southern edge of Çayırova district, railway route and waterways also located within One kilometer radius from the Kroman Chemical industry zone,
- One airport is located in the Darica district but this airport is beyond 5km radius from the Kroman Chemical industry,
- At Çayırova district, n=205 buildings are located within one kilometer radius within 1 km radius from the Chemetall Industry Zone,
- At Darica district, n=708 buildings are located within one kilometer radius within 1 km radius from the Kroman chemical industry zone,

Major Findings...

- 94% (n=103) of Industrial locations falls in the CORINE industry area,
- 6% (n=7) Industrial locations are located under urban fabric based on CORINE land distribution,
- Darica district total area square meter less by 5291 compare to Cayirova,
- At Darica <15 population high,
- n=1669 (29%) of facilities re located in the Cayirova district,
- n=4101 (71%) of facilities are located in the Darica district,
- 74% of Industries located at Cayirova,
- 26% of Industries located at Darica,
- 62% of apartments are located at Cayirova. They are far from Chemetall chemical industry but located 2km radius from Kroman chemical industry from Darica district,

Major Findings...

- 15% of Restaurants located in the Cayirova
- 85% of Restaurants located in DarIca,
- 19% of Bakery located in Cayirova,
- 81% of Bakery located in DarIca,
- 21% of Fast Food located at Cayirova,
- 79% of Fast Food located at DarIca,
- 100% shops located at DarIca,
- 10% of café located at Cayirova
- 90% Café located at Darica,
- 100% Butchery located at DarIca
- 100% Malls located at Darica
- 100% Dairy products located at DarIca,
- 100% Food Court located at DarIca,

Major Findings...

- 60% Schools are located at Darica district (a few found to be high risk area near the chemical industry,
- 40% of Schools are located at Cayirova (location all located low risk area)

Conclusion

- Based on the data from different scenarios, Darica districts was found to be a high risk place for all facilities from the point of Kroman chemical industry compare to Cayirova district,
- To strengthen the findings it would be also suggested to trace back all the surveillance data retrospectively since the establishment of the Chemical industry vs. households, schools, facilities confirmed reported cases based on hospital data, surveillance data to do data collection to see the time series analysis of data based on :
 - **Physical hazard (explosive, flammable...),**
 - **Health Hazard (Acute toxicity (dermal, oral and inhalation) and**
 - **Environmental hazard,**
 - **Acute respiratory,**
 - **Occupational asthma,**
 - **Skin irritation,**
 - **Physical**
 - **Biological**
 - **Chemical**
 - **Respiratory**
 - **Skin**
 - **Occupational Cancer**

Conclusion

- There is boundary discrepancy between the two different sources, i.e. from WHO/FMOH source and from the HDX source (shape file latest update shows 22 March 2021),
- Two Chemical industry factory functional including Kroman from Darcia and Chemetall from Cayirova. Ravago Petro Kimya Uretim A.S. is closed permanently,
- A total of n=297 (Donati) different facilities, located in the Cayirova and Darica district. Of Which within 1km radius, n=14 villages/towns/Donati are available, and n=1 villages/towns/Donati is available at Cayirova district near to Chemetall industry zone,
- A total of n=17 health facilities are located within the Cayirova and Darica districts. Of Which n=15 of them are Pharmacies, n=1 Hospital and n=1 Clinic,
- Only two pharmacies located within one kilometer radius from the point of Kroman Chemical industry zone, from Darica district,
- There is no facilities within one kilometer radius from the Chemetall chemical industry from Cayirova district,
- A total of n=22 schools are located in the territory of Cayirova and Darica district. A total of n=19 (88%) of Schools are available at Darica district and n=3(12%) are from Cayirova district,
- From n=22 schools, from the Kroman Chemical industry from Darica district, n=6 schools are located within 1km radius, n=5 with 2km radius, n=7 3km radius and n=4 beyond 3km radius,
- A total of n=4 industrial buildings are available within 1km radius from the Chemetall Chemical Industry zone from the Cayirova district,
- A total of n=5770 buildings are available in the territory of Cayirova and Darica district. Of Which n=1669 (29%) from Cayirova and n=4101 (71%) from Darica.
- Within 1km radius from the Chemetall chemical industry, a total of n=205 buildings observed from the Cayirova district, as a result, n=4 industrial buildings, n=1 factory and n=199 other buildings,
- Within 1km radius from the Kroman chemical industry, a total of n=708 buildings observed from the Darica district. Of which n=649 (91.7%) are categorized as buildings, n=4 (0.6%) Schools, n=52(7.3%) residential and the rest commercial and hotels.

Recommendations

- **By different km radius from the health facilities (could be Hospitals, Clinics, Pharmacies (based on the prescription data) if we can get data through surveillance/Disease Prevention and Control department/Epidemiology section from the Ministry of Health data for Çayırova and Darıca District based on Epidemic week based on the following parameters:**
 - 1. Province
 - 2. District
 - 3. Local geographic area
 - 4. Epidemic weeks
 - 5. Hospital reported confirmed cases of :
 - Skin disease
 - Inhalation disease
 - Occupational cancer
 - Respiratory
 - Eye damage
 - 6. Age
 - 7. Sex
 - 8. Occupation
 - 9. Date of onset
 - 10. Date of report
 - 11. Clinical diagnosis
 - 12. Final patient outcome
 - Acute
 - Oral
 - Dermal
 - Inhalation
 - Skin
 - Environmental
 - Aquatic
 - 3. Respiratory diseases,
 - 4. Skin diseases,
 - 5. Occupational Cancer,
 - Chemical
 - Physical
 - Biological agents,

Data source

- WHO and MOH
- HDX
 - [Welcome - Humanitarian Data Exchange \(humdata.org\)](http://humdata.org)
 - Point of interest
 - DONATI
 - Building
 - CORINE – EEA land distribution

HDX Data Source

The screenshot shows the HDX Data Source website in a web browser. The browser's address bar displays <https://data.humdata.org>. The website's header includes the HDX logo, a search bar labeled "Search Datasets", and navigation links for DATA, LOCATIONS, ORGANISATIONS, and QUICKLINKS. A red "ADD DATA" button is positioned on the right. Below the header, a large teal banner features the text "The Humanitarian Data Exchange" and the tagline "Find, share and use humanitarian data all in one place", accompanied by a "LEARN MORE" button. To the right, a "FIND DATA" section contains a search bar and statistics: 18,819 DATASETS, 254 LOCATIONS, and 1,472 SOURCES. Below this, an "ADD DATA" section shows icons for uploading data to the HDX and connecting with others via HDX Connect. A red banner at the bottom promotes the "Coronavirus COVID-19 Data Explorer". The Windows taskbar at the bottom shows the time as 2:28 PM on 8/12/2021.

OCHA Services

Data Responsibility for COVID-19 | ? FAQ | Log in | Sign up

HDX Search Datasets DATA | LOCATIONS | ORGANISATIONS | QUICKLINKS

ADD DATA

The Humanitarian Data Exchange

Find, share and use humanitarian data all in one place

[LEARN MORE](#)

FIND DATA

Search Datasets

18,819 DATASETS	254 LOCATIONS	1,472 SOURCES
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ADD DATA

Make your dataset available on HDX

HDX Connect: let others request your data

[Activate Windows](#)
Go to Settings to activate Windows.

[Coronavirus COVID-19 Data Explorer »](#)

Type here to search

Address

2:28 PM 8/12/2021

Report Status

- This is the first final Draft Report. Upon receiving comments, I will incorporate and will submit before August 21, 2021.
- I thank you very much Dr. Isabelle for your Support throughout the assignment.

Best Wishes, Negusu Worku GEBRMICHAEL

August 12, 2021